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RICOSLAM: LOCALIZATION AND MAPPING IN DYNAMIC ENVIRONMENTS

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Doctoral Program in Electrical and Computer Engineering

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Resumo

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Abstract

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Acknowledgments

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Next, a special thanks to my colleagues from CRIIS, including from the laboratories iiLab – Industry and Innovation Laboratory and TRIBElab – Laboratory of Robotics and IoT for Smart Precision Agriculture and Forestry. Although my research is related to the intralogistics industry and more related to iiLab's scope and mission, I began my PhD journey working side by side with TRIBElab, as the lab is located on the campus of the Faculty of Engineering, University of Porto (FEUP). This experience was excellent in terms of the team environment and spirit, while also providing the opportunity to interact daily with my mates from the master's degree, namely André and Daniel. Additionally, the TRIBElab leader, Filipe Santos, was one of the most important people who helped me decide that staying at INESC TEC would be the best choice to continue my journey and pursue a PhD degree, given the interaction with real-world applications and the opportunity to move beyond simulators and the theoretical world. So, a special thanks to him for that support, and to André and Daniel, two close companions and friends who supported me greatly throughout my PhD. After I moved to the iiLab facilities, it has been a fantastic journey, marked by the creation of new team partnerships and demonstrating that working side by side with colleagues from the same application field and different research areas can result in fascinating research experiences. A special thanks to my team leader Héber, also the co-supervisor of my PhD Thesis, Prof. Manuel Silva, for his guidance on the GreenAuto project directly linked with my PhD research and all the support given by him and the current and past CRIIS coordination team, to all my teammates of the Mobile Robotics Development Team (MRDT), and all my colleagues from iiLab that I interact daily with and has been a pleasure to work with them.

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Ricardo B. Sousa

“I don’t stop when I’m tired.
I stop when I’m done.”

David Goggins

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Abbreviations

3DGS 3D Gaussian Splatting. 36

ADRA AI, Data and Robotics Association. 3
AI Artificial Intelligence. 3, 18, 19
AMCL Adaptive Monte Carlo Localization. 51
AMR Autonomous Mobile Robot. 1–4, 11, 38
ARMA Auto-regressive Moving Average Model. 37, 40
ATE Absolute Trajectory Error. 30, 32, 35–37

BDVA Big Data Value Association. 3
BEV Bird-Eye-View. 32
BoW Bag-of-Words. 34
BPF Bayesian Persistance Filter. 37

CAD Computer-Aided Design. 31
CAGR Compound Annual Growth Rate. 1
CLAIRE Confederation of Laboratories for Artificial Intelligence Research in Europe. 3
CML Concurrent Mapping and Localization. 11
CNN Convolutional Neural Network. 34
CRF Conditional Random Field. 36
CRIIS Centre for Robotics in Industry and Intelligent Systems. 7

DBN Dynamic Bayesian Network. 13, 14
DBSCAN Density Based Spatial Clustering of Applications with Noise. 46
DDE Dynamic Data Extraction. 25
DE Data Extraction. 22, 23
DEL Dynamic Edge Link. 33
DFR-SLAM Dynamic visual Feature Removal SLAM. 32

EKF Extended Kalman Filter. 14, 15, 17, 18, 36, 46
ELLIS European Laboratory for Learning and Intelligent Systems. 3
EU European Union. 3
EurAI European Association for Artificial Intelligence. 3

FEUP Faculty of Engineering, University of Porto. 7, 24
FFT Fast Fourier Transform. 40
FOV Field-of-View. 4, 51–53, 55
FreMEn Frequency Map Enhancement. 37
FSH Feature Stability Histogram. 30

GTSAM Georgia Tech Smoothing and Mapping. 15

HMM Hidden Markov Model. 36, 37, 40

HOG Histogram of Oriented Gradients. 35

ICP Iterative Closest Point. 6, 13, 31, 33, 38
iiLab Industry and Innovation Laboratory. 7
IMU Inertial Measurement Unit. 35
INESC TEC Institute for Systems and Computer Engineering, Technology and Science. 7, 24
iSAM incremental Smoothing And Mapping. 15
ISDOR Incremental Static-referenced Dynamic Object Removal. 35

LC-CRF Long-term Consistent Conditional Random Field. 32
LiDAR Light Detection And Ranging. 2, 13, 32–35, 37, 38
LOAM Lidar Odometry and Mapping. 34
LTD Long-Term Dynamics. 35
LTM Long-Term Memory. 2, 30, 34, 39
LTS-NET Long-Term Stability NETwork. 38

MCL Monte Carlo Localization. 31, 37, 38
MLE Maximum Likelihood Estimation. 16

NDT Normal Distribution Transform. 13, 33, 38
NeRF Neural Radiance Field. 36
NICP Normal Iterative Closest Point. 13
NN Nearest Neighbor. 32

ORB Oriented FAST and Rotated BRIEF. 35

PCA Principle Component Analysis. 46
PPS Product, Processes, or Services. 4, 7
PRISMA Preferred Reporting Items for Systematic reviews and Meta-Analyses. 17, 19–21, 39

RANSAC RANdom SAmple Consensus. 46, 49
RBPF Rao-Blackwellized Particle Filter. 13, 14, 38
RMSE Root Mean Square Error. 34–36, 48, 49
RNN Recurrent Neural Network. 34, 35
ROS Robot Operating System. 8, 9, 16, 46, 51, 54, 55
RPE Relative Pose Error. 36

SDF Signed Distance Field. 31
SDG Sustainable Development Goal. 7
SGC-VSLAM Semantic and Geometric Contraints Visual SLAM. 35
SIVO Semantically Informed Visual Odometry. 34
SLAM Simultaneous Localization and Mapping. 1, 2, 4–9, 11–19, 21, 25, 34, 36, 38–41, 51, 55
SM Sensory Memory. 2, 30, 39
SRIDA Strategic Research, Innovation, and Deployment Agenda. 3
STD Short-Term Dynamics. 35
STM Short-Term Memory. 2, 30, 39
SURF Speeded Up Robust Features. 54
SVD Singular Value Decomposition. 46

TORO Tree-based netwORk Optimizer. 15

UN United Nations. 7

w.r.t. with respect to. 5, 7
WP Work Package. 3, 4, 7

Chapter 1

Introduction

This research, *RicoSLAM: localization and mapping in dynamic environments*, aims to advance the current state of the art in Simultaneous Localization and Mapping (SLAM). This work focuses on improving localization and mapping performance when performing SLAM in dynamic environments, using wheeled odometry and 2D laser scanner data from sensors onboard industrial ground mobile robots. SLAM in dynamic environments is closely tied to lifelong SLAM and long-term operation of mobile robots, enabling robots to maintain accurate localization and mapping over extended periods while adapting to continuous environmental changes.

This chapter is divided into seven sections. First, Section 1.1 contextualizes this research work and describes the primary challenges related to localization and mapping in dynamic environments. Section 1.2 outlines the key motivations and reasons for undertaking this research. Next, Section 1.3 introduces the problem that this PhD Thesis seeks to address. Section 1.4 presents the main research questions and hypotheses that guided the research work. Then, Section 1.5 defines the scope and objectives of the PhD Thesis. Section 1.6 highlights the unique contributions this research makes to the literature, emphasizing its novelty and impact. Lastly, Section 1.7 outlines the document's organization. A final note is provided regarding the funding that supported the work carried out throughout the PhD Thesis.

1.1 Context

Nowadays, Autonomous Mobile Robots (AMRs) are being increasingly deployed in several industries, including logistics, retail, warehousing, automotive and aerospace manufacturing, agriculture, and healthcare. AMRs may automate tasks such as material transport, automated inspection, and security monitoring. By improving operational efficiency, flexibility and productivity, AMRs help companies achieve higher efficiencies and reduce production costs. Consequently, these improvements contribute to the growth of the global economy and increase the worldwide interest in the AMR market. According to a study by ABI Research, the global robotics market is estimated to be roughly US$50 billion in 2025, growing at a Compound Annual Growth Rate (CAGR) of about 14% to reach approximately US$111 billion by 2030. This market growth is driven by labour workforce shortages, supply chain reformation, and high turnover rates, with the concurrent increasing consumer demand. Mobile robots in particular can mitigate these challenges by expediting the loading/unloading of trucks, automating inventory management, and performing last-mile delivery services [1].

An AMR must interpret its surroundings by extracting meaningful data from its sensors to localize itself within the environment, enabling autonomous navigation [2]. SLAM addresses this perception and localization challenge by concurrently incorporating the robot state estimation into the mapping process to build a representation of the environment as perceived by the robot. The state is typically described by the robot's pose (position and orientation), although other variables, such as velocity and calibration parameters, may also be included in the state. As for the environment representation, the mapping process involves creating a map incrementally, built as the robot explores unknown areas or refines it through passages in known locations, to describe the environment in which the robot operates. The map representation serves multiple purposes, such as path planning, providing a visual reference for human operators, while allowing the robot to reset its localization error when revisiting known areas [3].

In a static scene, the robot would only need to map once because the map and sensor data would remain consistent with the environment throughout time. However, industrial and intralogistics locations, agricultural sites, warehouses, and service-oriented applications, such as shopping centers and healthcare institutions, have inherently moving elements in the scene (humans, objects), reconfiguration of the environment (reorganization of machines, patient beds), and appearance variations (lighting, weather, seasonal, or daytime changes). The deployment of AMRs in these varying conditions are a challenge for the SLAM system, where the system should decide how (consider the most current state, the evolution of the environment dynamics, or only the most static changes?) and when to update the map (when the variations occur, or only after a certain time?). This challenge is also known as the stability-plasticity dilemma, where a lifelong SLAM system should both adapt to new environment changes and preserve old states over time [4]. Furthermore, the map would grow indefinitely when gathering new information from the environment without any map management system. This ever-growing problem poses another challenge for the SLAM system in the long-term due to the limited computational resources of the mapping vehicle. Indeed, the map size should be dependent on the environment area and not on the trajectory length of the mapping process nor operation time, only growing when the robot would explore unknown locations [5].

Currently, most methods rely on conventional SLAM map representations and then augment them with mechanisms to reason about dynamics. A recurring strategy for handling scene dynamics is to organize information into Sensory Memory (SM), Short-Term Memory (STM), and/or Long-Term Memory (LTM) [4, 6–9]. Another common trend is to partition the map into static or permanent, semi-static or movable, and dynamic layers, where static structures remain permanent over time, semi-static ones change only between visits at the same location, and dynamic layers represent sensor observations that move or are highly unstable in the environment [10–20]. Moreover, identification of scene dynamics is typically made through data consistency analysis, evaluating scan-to-map or scan-to-scan mismatches [6, 10, 11, 15, 17, 21–29], reprojection errors [19, 30–33], motion inconsistencies in tracked features [8, 19, 31, 33–36], ray-tracing discrepancies [4, 9, 11, 21, 22, 25, 37–39], and multi-session inconsistencies [24, 29, 38]. Clustering may improve dynamics identification through grouping nearby dynamic observations [7, 16, 28, 35, 38, 40, 41]. Object detection and semantic segmentation, typically from visual or 3D Light Detection And Ranging (LiDAR) data, may facilitate the identification of scene dynamics, especially when neural networks are used to classify objects in sensor observations, considering some of them dynamic, movable, or static [8, 12–14, 18–20, 33–36, 42–47]. Also, dynamic prediction may infer whether a region of the environment is stable, periodically occupied, or likely to change, to support more robust localization systems or even path-planning approaches that avoid highly dynamic areas [15, 16, 48–52]. Still, most methods rely primarily on static maps, excluding dynamic elements from the pose estimation process, and rarely maintain a continuously updated map representation.

1.2 Motivation

In May 2021, a joint partnership between major European organizations – Big Data Value Association (BDVA), Confederation of Laboratories for Artificial Intelligence Research in Europe (CLAIRE), European Laboratory for Learning and Intelligent Systems (ELLIS), European Association for Artificial Intelligence (EurAI), and euRobotics – founded the AI, Data and Robotics Association (ADRA), formed from the European Partnerships in Cluster 4 (digital, industry, and space) in Horizon Europe. Since then, ADRA, in collaboration with the European Commission and other community members, has worked on the Strategic Research, Innovation, and Deployment Agenda (SRIDA) [53]. The most recent version, published in February 2024, reflects on European strategies and goals that Artificial Intelligence (AI), data, and robotics can address on global challenges. This version outlines the ADRA's Vision 2030, the global challenges and primary strategic objectives that the ADRA partnership seeks to address, major trends and gaps, as well as recommendations for the European Union (EU)'s 2025–2027 strategic plan in alignment with the SRIDA.

SRIDA [53] identifies the need for a new wave of AI, data, and robotics methods that can tackle complex societal and industrial challenges. Driven by the accelerating progress of AMRs, these challenges could lead to the development of self-supervised robot perception and autonomous control mechanisms, enabling predictive and safe robot operation without human intervention. Moreover, SRIDA acknowledges that autonomy in robotics is currently constrained by short-term, localized environments with controlled uncertainty. There is an urgent need for higher levels of autonomy, particularly in complex, unknown, and dynamic environments. Indeed, these environments often include unanticipated scenarios, which robots must be capable of handling and adapting to. Consequently, robots may present significant risks to both people and the environment, underscoring the need for reliable and trustworthy autonomous operations. As a result, SRIDA motivates research on improving localization and mapping processes for industrial mobile robots within dynamic environments, aiming to increase their levels of autonomy. This expansion of the range of scenarios where robots can operate without human intervention contributes to the long-term operation and reliable, lifelong autonomy of AMRs.

Furthermore, the logistics industry has high dynamic environments with constantly shifting elements, such as pallets, boxes, workers, moving AMRs (base platforms, rack and cart lifting robots, car pulling AMRs, moving pallet platforms, conveyor robots, mobile manipulators, pallet jack AMRs, tugger robots to tow carts like a convoy, and autonomous forklifts [54]), and manually operated mobile machines. Additionally, logistics warehouse environments may have featureless characteristics that make it difficult for AMRs to localize within the environment over time. One situation that is particularly critical to address in dynamic logistics environments involving robots' localization is cases involving random slotting warehouse strategies, which optimize space and enhance efficiency in modern operations. In random slotting, products are stored in any available space in the warehouse based on their characteristics (weight and size), thereby reducing replenishment time and increasing warehouse efficiency [55]. Subsequently, the interpretation and processing of dynamic changes in the environment by AMRs would enable the maintenance of an up-to-date map representation of the robots' surroundings in logistics environments. An up-to-date representation would allow the use of dynamic elements in the scene that are temporarily static, thereby improving the robot localization estimation in the environment.

Finally, the Work Package (WP) 10 of the *GreenAuto – Green Innovation for the Automotive Industry* [56] project aims to develop technologies that reduce the production costs of electric vehicles, which are currently higher than those of traditional internal combustion engine alternatives. The use of autonomous logistic platforms on the factory floor, which either directly or indirectly assist in the production process, is already well established. However, there is significant room for improvement, particularly in extending the applicability of those systems to less structured environments. WP10 proposes the development of a navigation system for AMRs that is robust in complex environments. One of the key deliverables from WP10 is the Product, Processes, or Services (PPS) 18, a navigation system for mobile robots designed to be both robust and efficient in automotive shop floor settings. Consequently, PPS18 motivates the research and development of localization and mapping algorithms that account for dynamic elements in the environment, aiming to enhance the robustness of the final autonomous system.

1.3 Problem Statement

As discussed in Section 1.1, a map representation in static environments does not require updates, as it remains consistent with the surrounding environment that has been explored. However, most use cases for AMRs involve dynamic environments with moving elements or reconfigurations that affect the consistency of the map over time. Consequently, the system must determine how and when to update the map, a challenge known as the stability-plasticity dilemma. In this context, a long-term SLAM system must adapt to new changes while preserving older features, particularly when the environment reverts to previous states [4].

This PhD Thesis addresses the problem of localizing a mobile robot within dynamic indoor environments. This problem is directly related to SLAM, given the need to update the map concurrently with localization upon dynamic changes in the robot's surroundings. Furthermore, this research is not limited to a specific type of dynamics, for example, only detecting semi-static dynamics – changes between running sessions, *i.e.*, changes occurring outside of the sensors' Field-of-View (FOV) at different time instants that, when observed again by the sensors, appear static but elements were effectively moved in the scene – or only dealing with highly dynamic elements (*e.g.*, moving persons or objects). Instead, this thesis studies the detection of dynamic changes by comparing the current observation with a local reference map representation.

In terms of the sensor setup, this PhD Thesis assumes that the robot is equipped with at least one 2D laser scanner and, optionally, wheeled odometry data. This sensor setup is commonly found in AMRs within the logistics industry, where wheeled odometry is typically estimated by counting the ticks from the motor encoders while taking into account the robot's kinematic model [57]. In addition, AMRs often use 2D lasers not only for navigation but also as safety devices, in compliance with the ISO 10218-1:2025 [58] and ANSI/RIA R15.06-2025 [59] standards, which ensure a safe stopping distance and help prevent collisions with obstacles. Also, the usage of 2D lasers as safety devices allows AMRs to be compliant with ISO 3691-4:2023 [60], where, *e.g.*, an AMR must detect persons by passing on the two following tests: detecting a cylinder of 200 mm diameter and 600 mm length placed horizontally on the ground; and detect another cylinder of 70 mm diameter and 400 mm length set vertically. Consequently, this thesis focuses solely on range data.

1.4 Research Questions and Hypotheses

In response to the problem outlined in Section 1.3, this thesis proposes some research questions aimed at addressing and solving the identified problem. For each of these research questions, we formulate hypotheses that guided the investigation toward feasible solutions.

Research Question 1. ***How to improve the mapping results when performing SLAM with a ground mobile robot in a dynamic indoor environment?***

In dynamic environments, SLAM algorithms can be affected by the presence of moving objects and changing surroundings. One key contributing factor to reduced SLAM accuracy is the inability of traditional static map representations to stay updated with real-time changes in the environment, leading to mismatches between current sensor observations and previously mapped data. As a result, the correspondence between these observations and the reference environment representation becomes weaker, leading to degraded map quality. Another issue arises from the growing amount of information to process over time. As the robot moves, the accumulation of data from sensors can lead to redundant information, especially in environments where sensor noise, matching errors, or outlier observations occur frequently. Consequently, the appearance of "fat" walls in the map representation obtained from SLAM can occur. Another consequence of this issue is that, while the map may appear fine locally, distortions and inaccuracies can accumulate at the global level, leading to inconsistencies between the map representation and the real environment itself.

Given the previous challenges, we propose research hypotheses aimed at improving mapping results during SLAM in dynamic indoor environments. The primary focus is on removing outdated or redundant observations from the environment representation to reduce the noise introduced by both the sensor and the matching process, continuously updating the map with the latest sensor observations to increase the inliers matching ratio, while exploring concurrently offline global bundle adjustment techniques to remove distortions from the global map.

Hypothesis 1.1. *Exploring the application of ray tracing techniques to update local map representations online continuously, based on the latest sensor observations. These techniques can help determine whether a sensor observation corresponds to a new point or an update to an outdated one, depending on whether the latest ranging data is closer than the existing map point or if the sensor observation is passing through an existing feature. Also, the continuous online update of local maps based on the latest observations may improve the online estimation for the robot pose and its stability during SLAM, by increasing the inliers ratio of the sensor observation alignment with respect to (w.r.t.) local map.*

Hypothesis 1.2. *Researching techniques for robot relocalization on existing graphs representations, avoiding the addition of redundant data to the map. Linking graph growth to the spatial extent of the environment explored, this approach could help avoid unnecessary graph expansion, keeping the graph size manageable and improving computational efficiency without the graph size being dependent on the operation time.*

Hypothesis 1.3. *Investigating the offline removal of dynamic points using ray tracing techniques during the generation of the environment representation. This representation could be leveraged by other algorithms for more accurate mobile robot localization by excluding transient or moving objects from the map.*

Hypothesis 1.4. *Studying offline global map refinement techniques, removing dynamic points in the process, while trying to remove map distortions from the online-based map representation.*

Research Question 2. ***How to integrate environment dynamics in the localization process when performing SLAM with a ground mobile robot in a dynamic indoor environment?***

The localization process in SLAM is also dependent on the quality of the sensor data, especially when dynamic elements are present in the environment. Moving objects, such as people, vehicles, or machinery, can introduce significant challenges during the data registration step. When these dynamic observations are considered during data registration without any outliers rejection technique, could introduce errors on pose estimation, potentially causing the robot's localization to drift or even fail entirely. In order to mitigate the effects of environment dynamics in the localization process, we explore several strategies formulated by the following research hypotheses. These hypotheses focus on extracting geometric features from the 2D laser scanner data, using robust kernels during the data registration process to reduce the influence of outliers while applying distance maps to avoid the traditional association step to speed up the data registration process. Additionally, implementing local map strategies to continuously update the local environment representation with the latest sensor observations can help maintain a more accurate knowledge of the robot's surroundings, improving localization stability in highly dynamic environments.

Hypothesis 2.1. *Exploring the use of geometric feature extraction, such as corners or lines, from 2D laser scanner data for localization purposes, rather than relying solely on raw ranging data. This approach may help filter out irrelevant data, providing more meaningful and stable features for localization, even in dynamic environments.*

Hypothesis 2.2. *Researching the use of distance map data structures for 2D point cloud registration. Distance maps can provide access to the nearest-neighbor point distance, enabling an Iterative Closest Point (ICP) [61]-like registration process without the need for the traditional association step. By applying robust kernels, this approach can improve the handling of dynamic environment point observations by considering them as outliers in the alignment optimization process or assigning them a lower weight.*

Hypothesis 2.3. *Exploring local map strategies to continuously update the local environment representation with the latest sensor observations. Unlike scan-based localization, which may be more prone to false positives or negatives, local maps facilitate loop closure detection through a scan-to-map alignment rather than scan-to-scan approaches that consider singular scans. Moreover, local maps can enhance the localization stability of mobile robots in highly dynamic environments, as they are continuously updated locally with the latest sensor observations, while allowing the incorporation of ray tracing-based techniques to detect dynamic elements and update the local representation accordingly.*

1.5 Scope and Objectives

Given the problem statement in Section 1.3, alongside the research questions outlined in Section 1.4, and considering the current advancements and state of the art in localization and mapping in dynamic environments as discussed in Chapter 2, this research thesis aims to explore and advance developments in SLAM using wheeled odometry and 2D laser scanner data, particularly focused on indoor industrial environments. We assume that the plant floor is planar, allowing just a 2D pose estimation to localize the robot in the environment.

In order to achieve the thesis's main goal, the following efforts were undertaken:

- modifications to existing mobile platforms to enable data acquisition for the purposes of this PhD Thesis;
- development of a line fitting-based corner-like feature extraction algorithm to capture corners geometric information from the environment on 2D laser scanners data;
- 2D point cloud data registration algorithms based on unsigned distance map data structures, proposing both point-to-point and point-to-plane error metrics;

Figure 1.1: The United Nations (UN) Sustainable Development Goals (SDGs) [64].

- a full distance map-based graph SLAM framework, where the front-end estimates the robot pose w.r.t. a local reference representation (*e.g.*, singular scan or a local map), generating odometry and loop closure constraints, while the back-end of the framework implements a pose graph optimization using the `srrg2_solver` [62];
- ray tracing-based techniques to deal with dynamic observations on the 2D point cloud map representation of the environment;
- offline global bundle adjustment formulation based on distance maps to try to remove distortions from the global map.

Furthermore, while the degree-granting institution is the Faculty of Engineering, University of Porto (FEUP), the host institution of this PhD Thesis is INESC TEC – Institute for Systems and Computer Engineering, Technology and Science. Specifically, all work conducted in this thesis was performed at iiLab – Industry and Innovation Laboratory, within CRIIS – Centre for Robotics in Industry and Intelligent Systems from INESC TEC. Throughout the PhD journey, the research was executed alongside PPS18 – *3D navigation system for mobile robotic equipment*, part of WP10 – *automated logistics for the automotive industry*, within the *GreenAuto Agenda: Green Innovation for the Automotive Industry* [56], as a contribution for the long-term SLAM research topic. This PPS involves several partners, including FEUP, INESC TEC, and the leading partner, Flowbotic Mobile Systems [63], responsible for the development of an autonomous forklift for the automotive industry.

Within the scope of the 2030 Agenda for sustainable development (see Figure 1.1), agreed by 195 nations in September 2015 [64], this research work falls under the following Sustainable Development Goals (SDGs):

- SDG8: Decent work and economic growth;
- SDG9: Industry, innovation and infrastructure;
- SDG12: Responsible Consumption and Production.

1.6 Contributions

This doctoral research advances localization and mapping for mobile robots in dynamic industrial environments by combining state of the art research, experimental evaluation, and system-level integration. The work resulted in peer-reviewed publications covering perception and feature extraction from 2D laser scanners, multimodal sensing integration, localization in dynamic environments, and SLAM benchmarking. Beyond the scientific contributions, this thesis also includes the public release of a dataset to support reproducibility and benchmarking, as well as the development of open-source software.

1.6.1 Main contributions

[65] R.B. Sousa, H.M. Sobreira, and A.P. Moreira. "A systematic literature review on long-term localization and mapping for mobile robots". In: *Journal of Field Robotics* 40.5 (2023), pp. 1245–1322. DOI: 10.1002/rob.22170.

[66] R.B. Sousa, H.M. Sobreira, M.F. Silva, and A.P. Moreira. "Line fitting-based corner-like detector for 2D laser scanners data". In: *2024 10th International Conference on Automation, Robotics and Applications (ICARA)*. Athens, Greece, 2024, pp. 210–216. DOI: 10.1109/ICARA60736.2024.10553006.

[67] R.B. Sousa, H.M. Sobreira, J.G. Martins, P.G. Costa, M.F. Silva, and A.P. Moreira. "Integrating multimodal perception into ground mobile robots". In: *2025 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*. Funchal, Portugal, 2025, pp. 104–111. DOI: 10.1109/ICARSC65809.2025.10970176.

1.6.2 Complementary contributions

[68] J.D. Ribeiro, R.B. Sousa, J.G. Martins, A.S. Aguiar, F.N. dos Santos, and H.M. Sobreira. "Indoor benchmark of 3-D LiDAR SLAM at iilab – industry and innovation laboratory". In: *IEEE Access* 13 (2025), pp. 212421–212442. DOI: 10.1109/ACCESS.2025.3643753.

[69] M.S. Lopes, A. Cordeiro, R.B. Sousa, J.A. Beça, P. Costa, J.P.C. Souza, and M.F. Silva. "CARGO: container-unloading autonomous robot with generalized operation". In: *2026 12th International Conference on Automation, Robotics and Applications (ICARA)*. Istanbul, Turkey, 2026, pp. 1–6.

1.6.3 Public datasets

[70] J.D. Ribeiro, R.B. Sousa, J.G. Martins, A.S. Aguiar, F.N. dos Santos, and H.M. Sobreira. *IILABS 3D: iilab indoor LiDAR-based SLAM dataset*. iilab – industry and innovation lab: INESC TEC – Institute for Systems, Computer Engineering, Technology, and Science, 2025. DOI: 10.25747/VHNJ-WM80.

1.6.4 Open-source Software

- https://github.com/INESCTEC/inesctec_mrdt_line_fit_based_corner_detector_2d_laser: C++ implementation compatible with Robot Operating System (ROS) 1 of the line fitting-based corner-like detector for 2D lasers [66];
- https://github.com/INESCTEC/inesctec_mrdt_hangfa_discovery_q2: repository containing all the documentation (hardware, 3D modelling, electronics, network configuration, sensor setup) and code (firmware,

C++ nodes compatible with both ROS 1 and ROS 2) associated with the modifications made on the Hangfa Discovery Q2 omnidirectional platform [71] to integrate multimodal perception [67];
  - https://oss.inesctec.pt/inesctec_mrdt_hangfa_discovery_q2/: GitHub.io public website to help researchers replicate the multimodal perception integration discussed in [67].
- https://github.com/sousarbarb/slr-ltlm-mr: repository containing all the documentation, data, and Python scripts used in the scope of the systematic literature review on long-term localization and mapping [65];
- https://github.com/sousarbarb/hector_slam: added an offline node to the original Hector SLAM [72] open-source repository (https://github.com/tu-darmstadt-ros-pkg/hector_slam) to process multiple ROS bag files, generate the 2D occupancy grid map, and save the robot pose data into TUM format [73];
- https://github.com/sousarbarb/slam_gmapping: same addition made to the `hector_slam` ROS package for the GMapping [74] open-source implementation;
- https://github.com/sousarbarb/stage_ros: native ROS wrapper for the Stage Simulator [75–77], added compatibility with both ROS 1 and ROS 2 versions;
- https://gitlab.com/sousarbarb_srrg-software/srrg_system: added CMake compatibility with both ROS 1 and ROS 2 versions of the packages `srrg2_core`, `srrg2_qgl_viewport`, and `srrg2_solver` [62].

1.7 Document Structure

The remaining of this document is organized as follows (**[TODO: update lated the following text]**):

- **Chapter 2** presents the SLAM problem, covering scan matching, particle filters, and pose graph formulations. The chapter also reviews literature on long-term SLAM in dynamic environments, including map representation, map matching, dynamic object detection, and dynamics prediction;
- **Chapter 3** describes the various mobile platforms and datasets used in this PhD Thesis, including robots such as the 5dpo omnidirectional robot, the Avantgarde tricycle and Flowbotics differential autonomous forklifts, and also details the IILABS 3D and other datasets from the literature;
- **Chapter 4** presents a corner-like feature detection algorithm for 2D laser-scanner data and evaluates its geometric invariance to various viewpoints;
- **Chapter 5** explores the concept of using distance maps for data registration, discussing optimization algorithms, covariance estimation, and point-to-point and point-to-plane alignment error formulations;
- **Chapter 6** presents the research conducted in this PhD Thesis on SLAM in dynamic environments, focusing on techniques such as online SLAM, dynamics removal, and offline global bundle adjustment;
- **Chapter 7** summarizes the conclusions and contributions of this research and offers final remarks on future work, including potential areas for further exploration and involvement in the scientific community.

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Chapter 2

Literature Review

This chapter discusses the current literature on SLAM in dynamic environments. Our systematic literature review [65] of the broader topic of long-term localization and mapping (see Appendix A) analyzes 142 works covering appearance invariance, modeling the environment dynamics, map size management, multi-session, and computational topics such as parallel computing and timing efficiency. The review article [65] serves as the foundation for this chapter's discussion. While this chapter adopts a similar systematic methodology, the discussion focuses on the scope of this PhD Thesis, namely, how the literature addresses scene dynamics in localization and mapping processes. As a result, this chapter extends the discussion of dynamics modelling presented in the review article [65] and updates the results of the latest full search inquiry to scientific databases, yielding an analysis of 53 studies, compared with the 32 originally covered in the review.

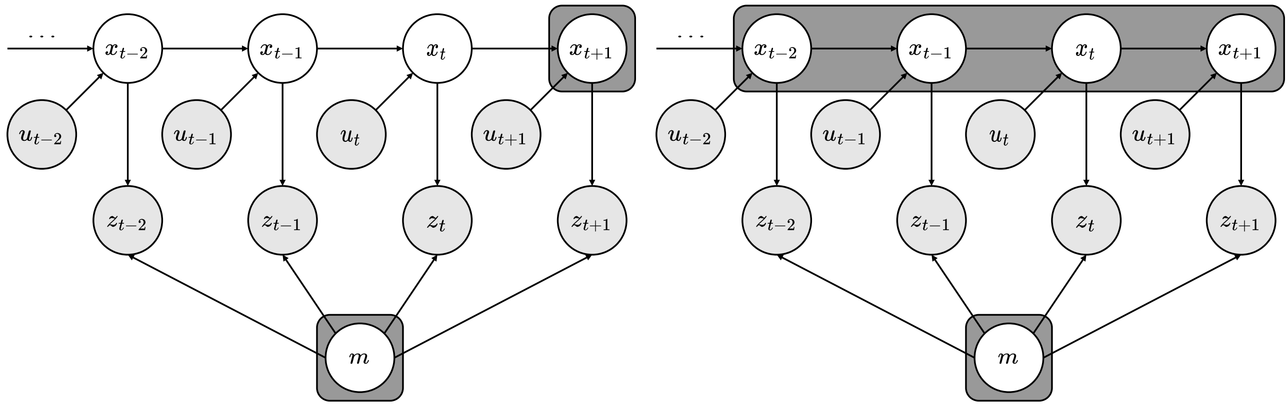
Hence, this chapter is divided into four sections. First, Section 2.1 formulates the traditional SLAM problem and presents three of its most common solutions: scan matching, particle filtering, and pose graph optimization. Next, Section 2.2 contextualizes the elaboration of a systematic literature review on long-term localization and mapping within the scope of this PhD Thesis, covering the main purpose of the study, the review methodology, and the results overview. Section 2.3 reviews the current literature on long-term localization and mapping in dynamic environments, focusing on how the literature addresses scene dynamics through map representation, map matching, dynamic object detection, and prediction modelling. Finally, Section 2.4 presents a summary of this chapter and the main conclusions.

2.1 Simultaneous Localization and Mapping (SLAM)

2.1.1 Problem introduction

SLAM is one of the most fundamental problems in robotics. This problem, also previously known as Concurrent Mapping and Localization (CML), may be formulated by Definition 2.1.1. When the robot does not have access to a representation of the environment nor know its pose, SLAM enables an AMR to build a map of its surroundings while simultaneously localizing itself. SLAM may also be used to build a map in applications assuming a static scene, even if the map is only continuously updated upon the mapping process [78].

Definition 2.1.1 ([2, p. 349])**.** *"...the robot must not only create a map, but it must also do this while moving and localizing to explore the environment. In the robotics community, this is often called the Simultaneous Localization and Mapping (SLAM) problem."*



(a) (b)

Figure 2.1: Graphical model of the Simultaneous Localization and Mapping (SLAM) problem: (a) online SLAM; (b) full SLAM [78].

Two main formulations define SLAM from a probabilistic perspective, also illustrated in Figure 2.1. The first is online SLAM, where the algorithm estimates the posterior over the momentary pose (x_t at a time instant t) along with the map (m), as defined in Equation (2.1), provided the sensor measurements ($z_{1:t}$) and the control variables ($u_{1:t}$). The online term is due to only involving the estimation of variables that persist at the current time instant t. Several algorithms incrementally implement online SLAM, *i.e.*, discarding past measurements and controls once they have been processed. The other probabilistic formulation is called full SLAM, given the goal to compute a posterior over the entire path ($x_{1:t}$) along with the map, as formulated in 2.2, instead of just the current pose (x_t). The difference between online and full SLAM is that the former results in an integration of the past poses from the full SLAM problem [78].

$$p\left(x_t, m \mid z_{1:t}, u_{1:t}\right) \tag{2.1}$$

$$p\left(x_{1:t}, m \mid z_{1:t}, u_{1:t}\right) \tag{2.2}$$

In the case of the algorithm also estimating correspondences between measurements and map landmarks, the explicit inclusion of the correspondence variables (c_t or $c_{1:t}$) in the estimation problem may be useful for reasoning if the perceived feature is the same one as previously detected and already represented in the map or if it is a new landmark. Consequently, Equation (2.3) and Equation (2.4) formulate the posterior estimation for online and full SLAM, respectively [78].

$$p\left(x_t, m, c_t \mid z_{1:t}, u_{1:t}\right) \tag{2.3}$$

$$p\left(x_{1:t}, m, c_{1:t} \mid z_{1:t}, u_{1:t}\right) \tag{2.4}$$

Although computing a full posterior captures all the information to be known about the map and the pose or the path, this estimation is practically unfeasible due to the high dimensionality of the continuous parameter space and the large number of discrete correspondence variables [78]. So, next, the discussion on scan matching, particle filter, and pose graph implementations of SLAM focuses on the practical aspects of performing continuous localization and mapping.

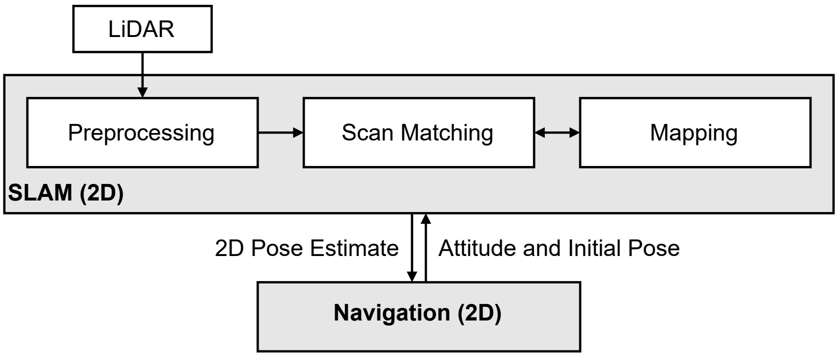


Figure 2.2: Block diagram of the mapping and navigation systems of HectorSLAM [72].

2.1.2 Scan matching

A basic approach for solving the SLAM problem is continuously matching the sensor scans to the map learned until the current instant. An example is HectorSLAM [72] algorithm, where the block diagram describing that system is shown in Figure 2.2. This algorithm is considered to be online SLAM due to only estimating the current pose and the map. Also, the method integrates incrementally the sensor readings to allow discarding those in subsequent iterations. Furthermore, HectorSLAM [72] aligns the beam endpoints of a 2D LiDAR with the current map using a matcher inspired by an alignment algorithm based on computer vision [79]. The matcher estimates the homogeneous transformation between the sensor scan and the map (*i.e.*, scan-to-map alignment) to update the latter with the latest scan, while also updating the current estimation of the robot pose. Although the matcher is specific to the HectorSLAM [72] algorithm, it does not invalidate using other matchers, such as the PerfectMatch [80], ICP [61], Normal Iterative Closest Point (NICP) [81], Normal Distribution Transform (NDT) [82], or others, to obtain the transformation between the 2D LiDAR scan and the map, allowing to perform SLAM with a scan matching methodology.

The main advantage of scan matching SLAM algorithms is the lower use of computational resources compared to more complex approaches. These algorithms do not require an exhaustive pose search or data association search with previous scans (the matching with all preceding scans is implicit in matching the current one with the map) [72]. However, as in the case of HectorSLAM [72], a pure scan matching approach does not implement loop closure (see Definition 2.1.2). This limitation prevents correcting the map when recognizing a previously mapped location, which means there is no reduction in the uncertainty of the map estimate, nor any correction for map distortions caused by incremental alignment errors.

Definition 2.1.2 ([2, p. 188])**.** *“The problem of location recognition is also called loop-detection, because a loop is a closed trajectory of a vehicle that returns to a previously-visited point. The problem of minimizing the error at the loop closure is instead called loop-closing.”*

2.1.3 Particle filter

The graphical model of the SLAM problem shown in Figure 2.1 is equivalent to a discrete Dynamic Bayesian Network (DBN) (see Definition 2.1.3). A sampling-based inference algorithm for DBNs is the particle filter. However, these filters suffer from the problem of dimensionality, scaling exponentially [83]. The assumption of conditional independence between map measurements allows using Rao-Blackwellized Particle Filters (RBPFs) to solve the SLAM problem. These filters use particles to represent the posterior probability over some variables (*e.g.*, the robot path, correspondence variables) and Gaussians (or other parametric probabilistic distribution functions) to represent all other variables (*e.g.*, the feature positions) [78]. This marginalization of some variables implemented by RBPFs improves the computational efficiency compared to standard particle filters, reducing the space size for the sampling step [83].

Definition 2.1.3 ([84]). *"A convenient way to describe this structure is via the Dynamic Bayesian Network (DBN)... A Bayesian network is a graphical model that describes a stochastic process as a directed graph. The graph has one node for each random variable in the process, and a directed edge (or arrow) between two nodes models a conditional dependence between them"*

An example of a RBPF-based SLAM is FastSLAM [85, 86]. This algorithm decomposes the SLAM problem into a robot localization one and the estimation of map landmarks conditioned on the robot pose estimate. The decomposition and factoring of the posterior probability ($p\left(x_t, m \mid z_{1:t}, u_{1:t}, c_{1:t}\right)$) followed by the FastSLAM [85, 86] algorithm is shown in Equation (2.5). The algorithm assumes the conditional independence of the features (m_n) given the robot path ($x_{1:t}^{[N]}$), *i.e.*, knowing the location of one feature does not add any additional information about the location of other features if the robot path is known. Moreover, Figure 2.3 represents the composition of particles implemented in FastSLAM. Each particle k contains an estimate of the robot path until the current time instant t and a set of Extended Kalman Filters (EKFs) with a mean ($\mu_j^{[k]}$) and covariance ($\Sigma_j^{[k]}$) of the estimation, one for each feature $j = 1..N$ represented in the map m. The basic steps of the FastSLAM [85, 86] algorithm are the following ones [78]:

- for each particle $k = 1..M$:
  1. **retrieval:** retrieve a pose $x_{t-1}^{[k]}$ from the particle set;
  2. **prediction:** sample a new pose $x_t^{[k]} \sim p\left(x_t \mid x_{t-1}^{[k]}, u_t\right)$ using the probabilistic motion model of the robot (the improved version, FastSLAM 2.0 [86], incorporates also the sensor measurements z_t^i in the prediction step);
  3. **measurement update:** for each observed feature obtained from a measurement z_t^i, identify the correspondence j in the particle map, and incorporate z_t^i into the corresponding EKF, updating the mean ($\mu_j^{[k]}$) and covariance ($\Sigma_j^{[k]}$) of the corresponding map feature accordingly;
  4. **importance weight:** compute the importance weight $w^{[k]}$ for the new particle.
- **resampling:** sample, with replacement, the M particles, where the probability of sampling being proportional to $w^{[k]}$, to incorporate the measurements z_t^i into the robot pose $x_t^{[k]}$ (due to only considering the most recent control u_t in the previous steps, only in FastSLAM 1.0 [85]).

$$\begin{aligned} p\left(x_t, m \mid z_{1:t}, u_{1:t}, c_{1:t}\right) &= p\left(x_t \mid z_{1:t}, u_{1:t}, c_{1:t}\right) p\left(m \mid x_t, z_{1:t}, c_{1:t}\right) = \\ &= p\left(x_t \mid z_{1:t}, u_{1:t}, c_{1:t}\right) \prod_{n=1}^{N} p\left(m_n \mid x_t, z_{1:t}, c_{1:t}\right) \end{aligned} \tag{2.5}$$

The adaptation of FastSLAM [85, 86] to occupancy grid-based SLAM is implemented by the GMapping [74] algorithm. This adaptation requires the modification of three modules relative to FastSLAM [85, 86]. First, GMapping [74] considers the occupancy grid as the map representation for the SLAM problem, where each particle contains a grid map. Next, the computation of a measurement likelihood ($p\left(z_t \mid x_t^{[k]}, m_{t-1}^{[k]}\right)$) uses the beam endpoint sensor model, characteristic of range-bearing sensors such as 2D lasers, instead of integrating the measures into an EKF as in FastSLAM [85, 86]. This model considers the individual scan beams independent from each other, using the distance between the beam endpoint and the closest obstacle in the map to that point to compute the beam's individual likelihood. Lastly, the update of each particle's map

	robot path	feature 1	feature 2		feature N
Particle $k=1$	$x_{1:t}^{[1]} = \{(x\ y\ \theta)^T\}_{1:t}^{[1]}$	$\mu_1^{[1]}, \Sigma_1^{[1]}$	$\mu_2^{[1]}, \Sigma_2^{[1]}$	$\cdots$	$\mu_N^{[1]}, \Sigma_N^{[1]}$
Particle $k=2$	$x_{1:t}^{[2]} = \{(x\ y\ \theta)^T\}_{1:t}^{[2]}$	$\mu_1^{[2]}, \Sigma_1^{[2]}$	$\mu_2^{[2]}, \Sigma_2^{[2]}$	$\cdots$	$\mu_N^{[2]}, \Sigma_N^{[2]}$
		$\vdots$			
Particle $k=M$	$x_{1:t}^{[M]} = \{(x\ y\ \theta)^T\}_{1:t}^{[M]}$	$\mu_1^{[M]}, \Sigma_1^{[M]}$	$\mu_2^{[M]}, \Sigma_2^{[M]}$	$\cdots$	$\mu_N^{[M]}, \Sigma_N^{[M]}$

Figure 2.3: Composition of particles in FastSLAM [85, 86] in terms of path estimate and the set of estimators and associated covariances for individual feature locations [78].

($p\left(m_t^{[k]} \mid z_t x_t^{[k]}, m_{t-1}^{[k]}\right)$) computes a new occupancy grid map. This update estimates the probability of each individual cell, assuming the independence of each other, using a binary Bayes filter and the inverse model of the sensor measurements [78].

A key advantage of particle filter-based SLAM is the decisions on data associations being made on a per-particle basis, instead of the most likely one as in scan matching or EKF approaches. This characteristic represents inherently multi-hypotheses associations in the filter, making it more robust to data association problems. Another advantage is particle filters being capable to model non-linear robot motions without requiring model approximations. However, a disadvantage of particle filters is relative to loop closure detection. Even though these filters are able to detect previously visited places, unlike scan matching approaches such as HectorSLAM [72], this place recognition ability is only maintained through the diversity in the particle sets, possibly maintaining larger sets and requiring more computational resources [78].

2.1.4 Pose graph

Another solution for the SLAM problem leveraging its graphical model (illustrated in Figure 2.1) is pose graph SLAM, also known as GraphSLAM. This optimization-based solution consists of two subsystems, as in filter-based approaches such as particle filters and EKF. In filter-based SLAM, the filters implement a prediction stage usually based on the vehicle's motion model, and then, the correction phase updates the predicted state by matching the sensor observation to the map. As for optimization-based solutions such as the posse graph formulation, a first module also known as the front-end is responsible to identify constraints (*e.g.*, odometry, loop closure, landmark observations) of the problem based on sensor data by finding correspondences between new observations and the current map. Then, a second subsystem known as the back-end computes or refines the poses of the graph and the map minimizing the coherence errors as a whole. This second module usually requires optimization algorithms such as Gauss-Newton or Levenberg-Marquardt for minimizing the objective function [87]. However, there are available in the literature optimization frameworks more specific for the pose graph formulation, namely, Tree-based netwORk Optimizer (TORO) [88], incremental Smoothing And Mapping (iSAM) [89, 90], Georgia Tech Smoothing and Mapping (GTSAM) [91, 92], g^2o [93], or `srrg2_solver` [62].

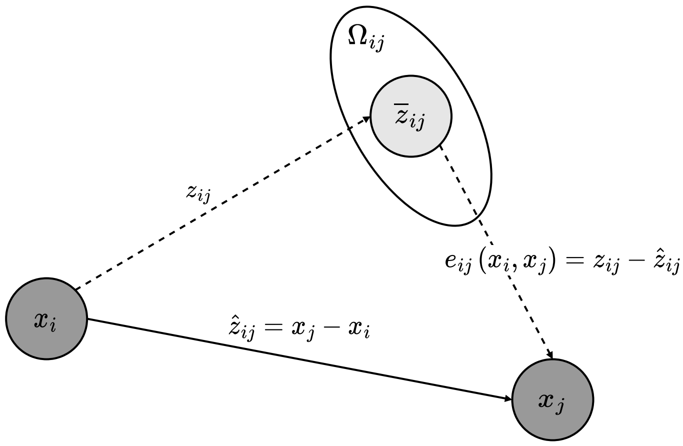


Figure 2.4: Edge connection in the graph formulation of the SLAM problem between vertexes x_i and x_j originated from a measurement z_{ij} [84].

Furthermore, the edge connection in GraphSLAM between vertexes x_i and x_j is illustrated in Figure 2.4. Considering z_{ij} as a virtual measurement obtained from raw sensor data between those two vertexes, let $\bar{z}_{ij}$ and Ω_{ij} be the mean and the information matrix of the virtual data, respectively. This virtual measurement actually represents the transformation that maximizes the overlap between the observations acquired from the pose nodes i and j. The log-likelihood l_{ij} (see Definition 2.1.4) of the measurement z_{ij} can be defined as in Equation (2.6), where $\hat{z}_{ij}(x_i, x_j)$ is the prediction of the virtual measurement given the configuration of the nodes x_i and x_j. Then, the configuration of nodes $x^* = (x_1, \ldots, x_T)^T$ can be found formulating a Maximum Likelihood Estimation (MLE) problem that minimizes the negative log-likelihood function $F(x)$ (see Equation (2.7)) of all the observations, seeking to solve the Equation (2.8) [84]. For exemplification purposes, two possible examples of virtual measurements are wheeled odometry and 2D laser scan matching data. Both cases would define the prediction of the measurement as $\hat{z}_{ij} = x_j - x_i$. The virtual measurement would be $z_{ij} = g(x_i, u_{ij})$ for wheeled odometry, where $g(x_i, u_{ij})$ represents the relative transformation between x_i and x_j estimated by the odometry estimator, while using the wheeled odometry data u_{ij} between those two nodes. As for the scan matching case, the measurement would be $z_{ij} = h(z_i, z_j)$ representing the matching between the laser scans z_i and z_j acquired from the nodes x_i and x_j, respectively.

Definition 2.1.4 ([94, p. 224]). *"...since the natural logarithm is a one-to-one function on $\mathbb{R}^+$, no information about θ is lost when the log-likelihood function $l(\theta) = \ln[p(\theta)]$ is used in place of $p(\theta)$. Moreover, for many probability models, it is easier to work with the log-likelihood function than with the likelihood function..."*

$$l_{ij} \propto [z_{ij} - \hat{z}_{ij}(x_i, x_j)]^T \Omega_{ij} [z_{ij} - \hat{z}_{ij}(x_i, x_j)] \tag{2.6}$$

$$F(x) = \sum_{i,j} e_{ij}^T \Omega_{ij} e_{ij} \tag{2.7}$$

$$x^* = \underset{x}{\operatorname{argmin}} F(x) \tag{2.8}$$

A pose graph SLAM implementation is the SLAM Toolbox framework [95], also available in the official navigation stack of the ROS [96] (in version 2). This toolbox is intended for 2D SLAM using a 2D laser scanner in conjunction with wheeled odometry obtained from the wheel encoders of the robot's motors. The optimization framework used for the pose graph back-end is the Ceres Solver [97]. Although SLAM Toolbox [95] states having lifelong mapping capabilities, these are only relative to being able to update existing maps and serialize the data for use in other mapping sessions, not for dynamic environments or appearance invariance.

Overall, a key difference between pose graph SLAM compared to other solutions for the SLAM problem, such as filter and scan matching approaches, is that the former has access to the full data when building the map, being able to apply improved linearization and data association techniques. Indeed, in GraphSLAM, all data can be used to linearize and calculate correspondences, *i.e.*, can revise past data association and linearize more than once [78]. This key difference is pointed as a possible reason to graph SLAM tending to give better results than filters, where the latter is more subject to linearization issues [87]. However, the pose graph formulation shows a linear growth over time in the graph's size requiring techniques for constraining the graph size, and thus, limit the computational resources required for continuously performing GraphSLAM [78].

2.2 Systematic Literature Review

Next, our systematic literature review on long-term localization and mapping for mobile robots [65] (see Appendix A) analyzes 142 works, following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) [98] guidelines. Instead of focusing on a single challenge, the review [65] covers several topics, including robustness to appearance changes, dynamics modelling, map sparsification to cope with the continuously increasing size of environment representations, multi-session mapping, and additional computational issues, such as parallel and distributed computing and memory management. The review [65] further provides a comparative analysis of the public datasets and experimental data adopted in the selected studies, focusing on scene conditions, sensing modalities, and their spatial and temporal properties. In addition, we made available a public GitHub repository[1] containing all the documentation and scripts used throughout the review process to enhance the repeatability and reproducibility of our review by the scientific community.

Although our review [65] covers a broader topic than the scope of this PhD Thesis, this section presents the systematic methodology followed in the review that underpins the discussion in Section 2.3. Lastly, this section examines the growing interest in long-term localization and mapping, based on the original search query used in the review, while adapting the query to investigate further the growing interest in handling scene dynamics.

2.2.1 Purpose of the study

The main studies reviewing the SLAM literature are presented in Table 2.1. In terms of introducing the problem formulation, the studies focus on explaining different frameworks for performing SLAM. Durrant-Whyte and Bailey [99, 100] discuss Bayesian-based probabilistic formulations, namely, EKF and the Rao-Blackwellized particle filter frameworks, also categorized as filtering approaches. In contrast, Grisetti *et al.* [84] explain in detail a smoothing formulation, the graph-based SLAM, which estimates the full trajectory of the robot from a set of sensor measurements. Thrun [101] introduces both probabilistic (EKF and particle filter) and smoothing (graph) formulations. Focusing on vision sensorization, Scaramuzza and Fraundorfer [102, 103] present an extensive tutorial on visual odometry for estimating relative motion from visual data, where the study discusses camera modeling and calibration, motion estimation, and feature matching. Yousif *et al.* [104] extend

[1] https://github.com/sousarbarb/slr-ltlm-mr

Table 2.1: Existent literature reviews, surveys, and tutorials on SLAM [65].

Year	Topic	Reference
2006	Probabilistic formulations (EKF, particle filter)	Durrant-Whyte and Bailey (2006) [99], Bailey and Durrant-Whyte (2006) [100]
2008	Probabilistic and pose graph formulations	Thrun (2008) [101]
2010	Graph SLAM	Grisetti *et al.* (2010) [84]
2011	Observability, convergence, consistency (feature-based SLAM)	Dissanayake *et al.* (2011) [105]
2012	Visual odometry	Scaramuzza and Fraundorfer (2011) [102], Fraundorfer and Scaramuzza (2012) [103]
2014	Underwater navigation and localization	Paull *et al.* (2014) [106]
2015	Visual SLAM	Yousif *et al.* (2015) [104]
2016	Observability, convergence, consistency (feature and graph-based SLAM)	Huang and Dissanayake (2016) [107]
	Multi-robot SLAM	Saeedi *et al.* (2016) [108]
	Visual place recognition	Lowry *et al.* (2016) [109]
	SLAM literature overview	Cadena *et al.* (2016) [3]
2017	Autonomous vehicles	Bresson *et al.* (2017) [87]
2018		Kuutti *et al.* (2018) [110]
2018	AI for long-term autonomy	Kunze *et al.* (2018) [111]
	Long-term sensorization	Zaffar *et al.* (2018) [112]
2020	Deep learning	Fayyad *et al.* (2020)[113]
	Multi-robot search and rescue	Queralta *et al.* (2020) [114]
2021	Self-driving vehicles	Badue *et al.* (2021) [115]
2022	Underground navigation	Ebadi *et al.* (2024) [116]

the previous visual odometry tutorial to include methodologies for the vision-based SLAM problem. Although the introductions mentioned here provide comprehensive explanations of the problem formulation itself, none of them focus the discussion on the possible long-term challenges of SLAM.

Moreover, other studies focus on theoretical aspects of the SLAM formulation. Dissanayake *et al.* [105] discuss the fundamental properties of SLAM: observability (whether the problem is solvable), convergence (whether the state uncertainty converges to a finite value), and consistency (whether the estimated state is unbiased). Huang and Dissanayake [107] extend the previous work [105] by providing an in-depth explanation of the fundamental properties and defining criteria for the performance evaluation of SLAM algorithms in terms of consistency, accuracy, and computational efficiency. In contrast to [105], which focuses mainly on filtering-based SLAM, Huang and Dissanayake [107] also discuss the properties for smoothing formulations. Both studies focus only on theoretical aspects, without addressing long-term localization and mapping.

In terms of surveys on SLAM trends, Cadena *et al.* [3] provide a broad overview of metric and semantic SLAM works. The work briefly discusses localization and mapping robustness in terms of loop closure validation and handling dynamic environments, and SLAM scalability in terms of pose graph sparsification and parallel and distributed computing. By contrast, Lowry *et al.* [109] focus on topological SLAM, providing a comprehensive review of visual place recognition. Although the study discusses the challenges of navigation across varying conditions, it is limited to vision sensors. Bresson *et al.* [87] survey trends in single- and multi-vehicle SLAM and in large-scale experiments for autonomous vehicles. Although the study compares the methods on accuracy, scalability, availability, recovery, map updateability, and scene dynamics, their study only refers to approaches composed of at least odometry and a mapping module, not discussing localization-only algorithms. Also, the discussion is more focused on loop closure and relocalization modules and on leveraging existing data, rather than on methodologies for addressing the long-term challenges of continuous SLAM.

Similar to [87], Kuutti *et al.* [110] and Badue *et al.* [115] analyze trends for self-driving vehicles. While Kuutti *et al.* [110] focus on sensorization and cooperative localization between vehicles, Badue *et al.* [115] concentrate on the architecture of autonomous driving systems. However, neither of those studies discuss the challenges of achieving long-term SLAM. Saeedi *et al.* [108] present a review of multi-robot SLAM. Even though the authors identify large-scale, dynamic environments, multi-session operations, and agent scalability as challenges for multi-robot SLAM, their study does not provide an overview of existing methodologies to address these problems. Paull *et al.* [106] and Ebadi *et al.* [116] also present overviews of the SLAM literature, but are more specific in their domains: the former focuses on autonomous underwater navigation, and the latter on SLAM in extreme underground environments. However, those two studies do not discuss long-term SLAM.

The surveys by Kunze *et al.* [111] and Zaffar *et al.* [112] focus on some challenges of long-term autonomy. Kunze *et al.* [111] provides a brief overview of how AI can enable the long-term operation of autonomous systems. Although the survey discusses challenges such as environments with varying appearance and the dynamics of moving elements, the discussion is limited to AI-related work. The work only briefly analyzes localization and mapping tasks, while also focusing on reasoning, human-robot interaction, and planning. As for Zaffar *et al.* [112], the study focuses solely on reviewing, discussing, and comparing sensors with respect to sensor lifetime, field operability, ease of replacement, and suitability for various environments.

Nevertheless, none of the studies presented in Table 2.1 describe their methodology for selecting the works for discussion. This limitation prevents other researchers from repeating the review process, *e.g.*, to update existing studies and maintain an up-to-date understanding of the current state of the art in SLAM.

As a result, *A systematic literature review on long-term localization and mapping for mobile robots* [65] (see Appendix A) was written in the scope of this PhD Thesis. The review follows the PRISMA [98] guidelines, defining a systematic methodology to ensure the repeatability of the selection and data extraction processes while allowing future updates over the discussion and the review's methodology. Furthermore, our systematic review does not focus on any specific strategy or publication time interval. These considerations improve the study's coverage of the long-term localization and mapping topic, compared to the existent SLAM surveys.

In summary, the following questions are the focus of our systematic review on long-term localization and mapping literature [65]:

- main challenges inherent to lifelong SLAM;
- main strategies for accomplishing long-term operations with mobile robots;
- public datasets commonly used for evaluating long-term localization and mapping algorithms;
- how the researchers evaluate the performance of autonomous systems in long-term operations.

2.2.2 Review methodology

A systematic literature review uses explicit, rigorous, and reproducible systematic methods to synthesize the findings of studies related to a particular research question, topic area, or phenomenon of interest. This type of review ensures the quality and trustworthiness of its findings by presenting a complete, organized, and summarized analysis of all works considered, while allowing others to replicate or update the review. The most common standard for performing a systematic review is the PRISMA statement [98]. Although PRISMA was

Table 2.2: Exclusion (E) criteria for the selection process of the systematic literature review [65].

E#	Criteria	Statement
E1	Index	Papers not indexed in a scientific publication venue
E2	Language	Full-text of the papers not published in English
E3	Subject Area	Papers not classified in the databases as Computer Science, Engineering, Mathematics, or Multidisciplinary
E4	Short Papers	Papers classified as short papers accordingly to the publication venue
E5	Gray, Secondary, and Tertiary Literature	Books, preprints, reports, reviews, thesis, ...
E6	Availability	Full-text of the papers not available in digital libraries
E7	Dataset	Papers that focus only on data collection
E8	Coverage	Papers using only odometry for localization
E9	Scope	Papers that focus on different and not related subjects

initially designed to evaluate the effects of health interventions, its methodology checklist items are general and applicable to other subject areas [117–120]. Thus, the methodology used in our systematic review on long-term localization and mapping, extensively discussed in Appendix A, follows the PRISMA guidelines. This methodology is composed of the following steps:

1. definition of eligibility criteria;
2. conceptualization of the search strategy;
3. selection process:
   (a) identification;
   (b) screening;
   (c) quality assessment.
4. data extraction;
5. synthesis and analysis of the included works.

First, we excluded eligible studies according to the exclusion criteria presented in Table 2.2. These criteria focus on the type of paper, availability, and relevance to the research topic of long-term localization and mapping. Studies must be published in scientific venues to ensure peer review. Only primary literature and English-language papers available in digital libraries are considered to ensure accessibility and reproducibility. Additionally, studies not categorized in the scientific databases under *Computer Science*, *Engineering*, *Mathematics*, or *Multidisciplinary* subject areas are excluded, as these fields are closely related to the research topic. We also excluded works that focus solely on datasets, as these are not considered original research, as well as odometry-only approaches, which are not viable for long-term operations. Lastly, we excluded papers that are not directly related to long-term localization and mapping.

Next, our search strategy for the systematic literature review aimed to identify relevant data sources and define the search fields and base string to consider during the process of obtaining the included records. The review considered multiple databases to increase the overall scientific coverage, namely, ACM [121], Dimensions [122], IEEE Xplore [123], INSPEC [124], Scopus [125], and Web of Science [126]. Furthermore, the *Title*, *Abstract*, and *Keywords* were the fields used to obtain search results and focus the search on the most relevant summary information. The *Keywords* field includes the author keywords, database-indexed terms, and, if available, uncontrolled keywords. Also, we did not impose any date restrictions to avoid missing important studies and to provide a more comprehensive discussion on the topic. Finally, we last performed the search

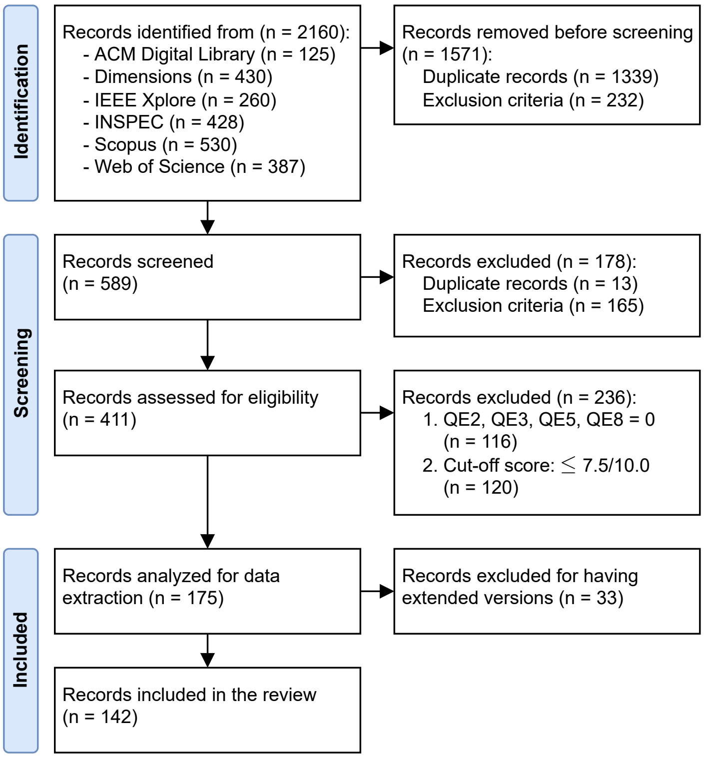


Figure 2.5: Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) flow diagram for the selection process [65].

on May 17, 2022, for the review. The search query used the following base string, combining terms related to robots, vehicles, localization, mapping, SLAM, and long-term operations, leveraging the asterisk ($\star$) wildcard to consider multiple synonyms.

```
(robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND
("long term" OR "life long" OR lifelong)
```

Figure 2.5 summarizes the selection process for the review. During the identification phase, the search strategy was applied to all data sources, using Parsifal [127] to identify duplicate records across databases. The 232 excluded records in this phase include works that do not meet criteria E4 (auxiliary Python scripts to identify short papers from BibTex bibliography data) and E7 (searching for the `dataset` term in the titles). We have gone through a double-check procedure to ensure these criteria are applied correctly. As a result, we screened 589 records, requiring us to read their titles and abstracts to determine eligibility for the quality assessment phase. Initially rejected papers were subject to a further assessment to validate the exclusion. This assessment included analyzing the experimental results and conclusions presented in the original works to either confirm or reverse the exclusion decision. Overall, we rejected 178 studies in the screening phase, leaving 411 eligible works for quality assessment.

The quality evaluation of the 411 eligible works followed the 8 Quality Evaluation (QE) criteria presented in Table 2.3. All of them are subjective criteria derived from the analysis of the eligible works, focusing on the detail and relevance of the proposed methodologies, experimental results, and long-term consideration. Considering the possible scores for the QE criteria in Table 2.3, each work can only have a maximum score of 10.0. After we evaluated the 411 eligible works, our first observation was that works with a non-detailed or appropriate methodology, results' discussion, or conclusions should not be included in the review. Another

Table 2.3: Quality Evaluation (QE) criteria and score range for the selection process of the systematic review [65].

QE#	Criteria	Score
QE1	Does the paper have an updated state of the art on long-term localization and mapping?	{0.0, 0.5, 1.0}
QE2	Is the methodology appropriate and detailed?	{0.0, 0.5, 1.0}
QE3	Does the methodology consider both localization and mapping problems?	{0.0, 1.0, 2.0}
QE4	Is the hardware and/or software used in the experiments detailed?	{0.0, 0.5, 1.0}
QE5	Does the paper presents any kind of long-term experimental results?	{0.0, 2.0}
QE6	Does the paper presents comparative results with other methods and/or ground-truth data?	{0.0, 1.0}
QE7	Does the work's implementation and/or the data used in the experiments are publicly available?	{0.0, 1.0}
QE8	Is the discussion of the results and conclusions appropriate and detailed?	{0.0, 0.5, 1.0}

observation is relative to rejecting works that do not consider either localization or mapping problems, or do not present any long-term experimental results. This rejection is due to the focus of the review on long-term localization and mapping. Furthermore, the quality assessment phase should include a cut-off score to filter out works with low quality scores. Therefore, we considered the following two reasons to reject a record in the quality assessment phase of the selection process, resulting in 175 studies for the data extraction phase:

1. QE2, QE3, QE5, QE8: reject works with a 0.0 (no compliance) score;
2. cut-off score: reject works with a score lower or equal to 7.5/10.0.

Finally, the data extraction process involved analyzing the eligible records from the quality assessment phase and extracting relevant information. In the scope of the systematic review on long-term localization and mapping, we considered the following Data Extraction (DE) items required for each record:

- [DE1] Long-term considerations – long-term factors the works consider in their proposed approach and experiments. Considering the knowledge obtained in the previous phases of the review's methodology, we considered the following factors for categorizing the included works:
  - appearance: varying conditions, appearance changes;
  - dynamics: environment dynamics, dynamic elements;
  - sparsity: map pruning, redundant data removal;
  - multi-session: map management;
  - computational: memory management, efficiency.
- [DE2] Localization – how the robot localizes itself and the type of localizer;
- [DE3] Mapping – type of the map;
- [DE4] Multi-robot – if the proposed methodologies consider multi-robot systems;
- [DE5] Execution mode – offline, online, if requires both, or if no information on this item;
- [DE6] Environment and domain – type of environment (indoor, outdoor) and domains (air, ground, water) tested with the proposed methodologies;
- [DE7] Sensory setup – which sensors considered in the methodologies;
- [DE8] Non-public experiments – if the authors performed experiments or tests with non-public data;
- [DE9] Ground-truth – how ground-truth for non-public data is obtained or its type, if available;
- [DE10] Distance and time characteristics – relative to the non-public experiments if available, as follows:

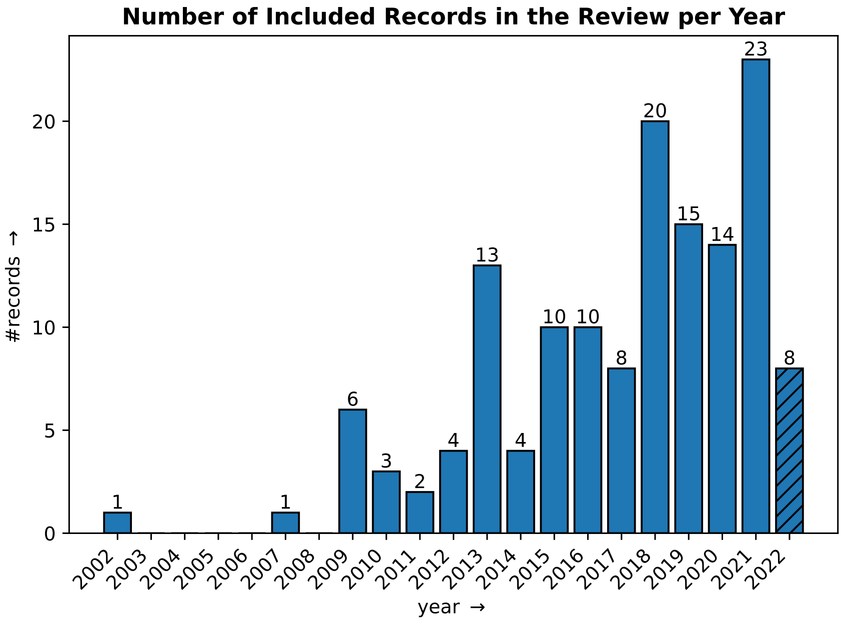


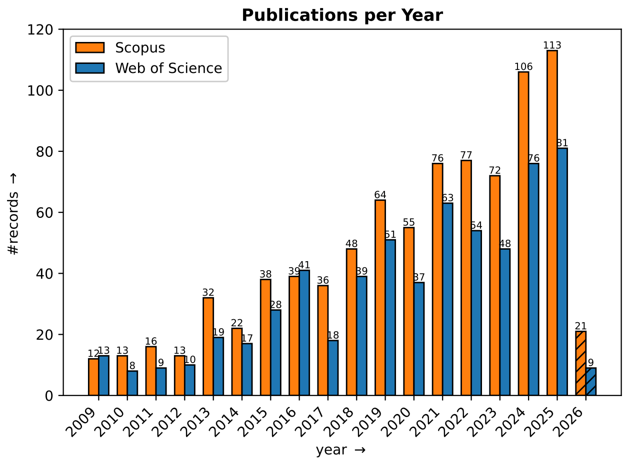
Figure 2.6: Evolution of published records per year considering the 142 included records in the systematic literature review on long-term localization and mapping. The time interval is between the smallest publication year found in the included records (2002) and the year of last full inquiry's date (2022). The latter is with a diagonal hatch pattern due to the fact that the last full inquiry did not consider the whole year [65].

  - total distance (km) of the non-public experiments;
  - path (km), in the case of repetitive paths;
  - total time (h) in terms of continuous operation;
  - time interval (day/week/month/year, or d/w/m/y) between the first and the last run.
- [*DE*11] Datasets – if and which public datasets are used in the experiments;
- [*DE*12] Evaluation metrics – which metrics are used for evaluation.

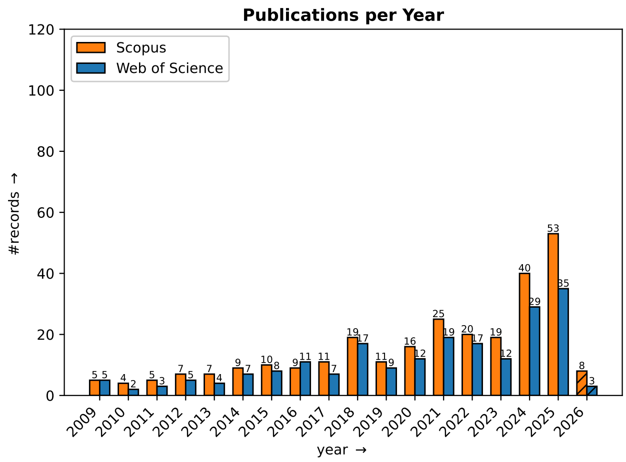
Although the data extraction phase typically does not exclude records, 33 of the 175 works analyzed had extended versions of the proposed methodologies, which we excluded to maintain the originality of the review. As a result, we included 142 original works for analysis and discussion in the systematic literature review study, representing 34.55% of the eligible records. The extracted data for these works is available in Appendix A.

2.2.3 Results overview

The publications per year evolution can be used to evaluate the relevance of the long-term localization and mapping topic. Figure 2.6 presents this evolution on the 142 included records in our systematic review. The final year, 2022, has its data with a diagonal hatch pattern, indicating that it does not represent a full year. Analyzing Figure 2.6, the long-term localization and mapping topic started to gained relevance in 2009, with 6 works, compared to only 2 in the previous years. From that year onwards, the graph shows an almost linear trend, reaching a maximum of 23 records in 2021, and already having 8 publications by May 17, 2022. This tendency indicates that long-term localization and mapping have gained interest over the years and, consequently, underscores the importance and relevance of the research topic to the scientific community.



(a)



(b)

Figure 2.7: Evolution of published records per year in Scopus [125] and Web of Science [126] databases on long-term localization and mapping: (a) `(robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong)`; (b) `(robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong) AND (dynamic* OR mov*)`. Date of the last full inquiry: February 13, 2026. The time interval is between 2009 and the year of last full inquiry's date (2026). The latter is with a diagonal hatch pattern due to the fact that the last full inquiry did not consider the whole year.

Although the scope of our systematic literature review [65] includes methodologies for handling the dynamicity of the scene, the original query did not focus on any particular challenge related to long-term localization and mapping. Figure 2.7 presents the evolution of publication year during the identification phase of a systematic review, using the *Title*, *Abstract*, and *Keywords* (authors' and indexed ones) as search fields, across two bibliographic databases: Scopus [125] and Web of Science [126]. The consideration of the previous two with INSPEC [124] could improve the coverage of the search results, as discussed in Appendix A, where the joint consideration of the three databases guarantees the identification of all 142 included records. However, the b-on[2] agreement provided by the Portuguese funding agency, FCT – Fundação para a Ciência e a Tecnologia, no longer has INSPEC [124] available to its participating entities, including FEUP and INESC TEC. Moreover, Figure 2.7a uses the exact inquiry string used in our systematic review. Figure 2.7b uses a different string to present the evolution of publications on the challenge of localization and mapping in dynamic environments, which aligns with the goals of this PhD Thesis. This scope reduction is accomplished by using the following search string:

```
(robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND
("long term" OR "life long" OR lifelong) AND (dynamic* OR mov*)
```

The part added to the original query (`dynamic* OR mov*`) aims to limit search results to works featuring dynamic or moving elements in the scene. Also, only works since 2009 were considered in the analysis, given that year was the one where the long-term localization and mapping started to gain traction in the literature.

When analyzing Figure 2.7a, the graph shows increasing interest among the scientific community in long-term localization and mapping. Since 2009, the graph shows a consistent upward trend, reaching a maximum of 113 and 81 published records in 2025 for Scopus [125] and Web of Science [126], respectively. The trend

[2] https://www.b-on.pt/

indicates that long-term SLAM is gaining prominence, underscoring its importance to the scientific community. This result conforms with the analysis of the year of publication for the 142 literature works included in our systematic review (see Figure 2.6).

Similarly, the graph shown in Figure 2.7b indicates growing interest in the topic of this PhD Thesis: localization and mapping in dynamic environments. From 2009 onwards, the topic has also gained relevance, achieving a maximum of 53 and 35 published works in 2025 for Scopus [125] and Web of Science [126], respectively. Although the overall number of publications related to dynamic environments is lower than for the broader long-term topic, the growing interest in addressing the challenges of localization and mapping in dynamic environments underscores the relevance of this PhD Thesis.

2.3 Long-Term SLAM in Dynamic Environments

In order to focus the literature discussion in this chapter on localization and mapping in dynamic environments, this section follows the same systematic methodology outlined in Section 2.2.2. The discussion builds on the analysis already made in the review article [65] on dynamics modelling. This analysis examines how literature handles scene dynamics through map representation, map matching, dynamic object detection, and predictive modelling. However, similarly to Section 2.2.3, the original search query is extended with `dynamic* OR mov*` to update the included works from the original date of the last full search inquiry in the scientific databases, May 17, 2022, to February 13, 2026, while focusing on works that handle scene dynamics. Also, this last full inquiry includes only Scopus [125] and Web of Science [126] search results to facilitate the update to the discussion. As a result, after completing all stages of the review methodology described in Appendix A and briefly presented in Section 2.2.2, this section discusses 53 works from the literature, including the 32 studies related to handling scene dynamics already discussed in the review article.

Table 2.4 presents the data extraction results for the 53 works identified in the latest full inquiry on dynamics modelling. The Dynamic Data Extraction (DDE) items differ from those presented in 2.2.2, as they are designed to capture meaningful information within the scope of localization and mapping in dynamic environments. Accordingly, the DDE items required for each of the 53 records are as follows:

- [*DDE*1] Mapping – mapping process and type of the map;
- [*DDE*2] Localization – how the robot localizes itself and the type of localizer;
- [*DDE*3] Environment and domain – type of environment (indoor, outdoor) and domains (air, ground, water) tested with the proposed methodologies;
- [*DDE*4] Sensory setup – which sensors considered in the methodologies;
- [*DDE*5] Dynamics detection – how dynamics identified from sensor observations;
- [*DDE*6] Dynamics labelling – classification of dynamic observations;
- [*DDE*7] Dynamics statistics – statistics formulated in the methodologies related to dynamics modelling;
- [*DDE*8] Dynamics prediction – methodologies to predict scene dynamics in the environment;
- [*DDE*9] Dynamics removal – how dynamics are handled in localization and mapping processes.

Next, complementing the information available in Table 2.4, the following analysis discusses how the 53 works included in this section address scene dynamics through map representation, map matching, dynamic object detection, and prediction modelling.

Table 2.4: Data extraction items retrieved from literature works related to long-term localization and mapping in dynamic environments.

DDE:	**1:**	**2:**	**3:**					**4:**	**5–9: Dynamics**				
Ref.	**Mapping**	**Localization**	**Indoor**	**Outdoor**	**Air**	**Ground**	**Water**	**Sensors**	**Detection**	**Labelling**	**Statistics**	**Prediction**	**Removal**
Biber and Duckett (2009) [4]	submap (2D point cloud, graph, 3DoF edges)	point cloud alignment (2D, 3DoF)	x			x		wheel odometry, laser (2D)	ray tracing, 1D robust statistics over dynamic sample sets (median, median of absolute differences), timescales	–	timescale (update policy)	–	dynamic sample set (timescale-based probability)
Dayoub *et al.* (2011) [6]	keyframe (graph, 6DoF edges)	feature matching (location)	x			x		camera (omni)	scan-to-map matching	sensory + short + long-term memory (SM, STM, LTM)	hit + miss observations	–	recall mechanism (forget unused features from LTM)
Walcott-Bryant *et al.* (2012) [10]	pose graph (graph, 2D point clouds, 3DoF edges; iSAM)	point cloud alignment (2D, 3DoF)	x			x		laser (2D)	scan-to-map matching	active + dynamic map, static + removed + added points	change score	–	disable node sectors, inactive nodes removal (all sectors turned off)
Bacca *et al.* (2013) [7]	keyframe (graph; TORO)	Bayesian (2D, 3DoF)	x			x		camera (omni), laser (2D)	k-means clustering	sensory + short + long-term memory (SM, STM, LTM)	feature stability histogram	–	STM pruning
Einhorn and Gross (2013) [128]	pose graph (graph, 2D/3D NDT; g^2o)	odometry	x			x		camera (mono, RGBD), laser (2D)	explicit free space mapping	–	–	–	explicit free space mapping
Tipaldi *et al.* (2013) [48]	grid (occupancy, 2D)	particle filter (2D, 3DoF)	x			x		laser (2D)	hidden Markov model (HMM)	–	mixing time of Markov chain	dynamic cells transition	–
Saarinen *et al.* (2013) [21]	3D NDT, grid (occupancy, 3D)	–	x			x		camera (RGBD), laser (3D)	ray tracing, scan-to-map matching (NDT-based)	–	–	–	ad-hoc sensor model
Pomerleau *et al.* (2014) [37]	point cloud (laser, 3D)	point cloud alignment (3D, 3DoF)		x		x		wheel odometry, laser (3D)	ray tracing approximation (sparse map cloud, kd-tree w/ spherical coordinates)	–	Bayesian (visibility, points proximity, motion patterns)	–	down-weight dynamic points
Einhorn and Gross (2015) [40]	pose graph (graph, 2D/3D NDT)	odometry	x			x		camera (mono, RGBD), laser (2D)	occupancy probability (short vs static / long-term)	dynamic cell clusters	static occupancy probability	–	–
Rapp *et al.* (2015) [49]	grid (occupancy, 2D)	particle filter (2D, 3DoF)		x		x		odometry, radar	semi-Markov chain extension	–	cell reliability (Lévy process)	expected retention time on cell state	–
Santos *et al.* (2016) [51]	spatio-temporal grid (occupancy, 3D)	–	x			x		camera (RGBD)	fast Fourier transform (FFT, FreMEn) (static vs periodic changes)	–	mean value, dominant frequencies	fast Fourier transform (FFT, FreMEn) (periodic changes)	–
An *et al.* (2016) [129]	pose graph (graph)	EKF (2D, 3DoF)	x			x		wheel odometry, camera (gray, mono), laser (2D)	feature matching	–	edge weight (dynamic edge link)	–	nodes removal (average edge weight threshold)
Biswas and Veloso (2017) [9]	feature (2D line segments), varying graphical network (VGN, Ceres)	maximum likelihood estimation (MLE)	x	x		x		wheel odometry, laser (2D), camera (RGBD)	feature ray tracing	dynamic (only for obstacle avoidance) + short (movable) + long-term features (DF, STF, LTF)	–	–	dynamic features removal (localization)
Krajník *et al.* (2017) [52]	grid (occupancy, 2D/3D)	–	x	x		x		camera (RGBD), laser (2D)	fast Fourier transform (FFT, FreMEn) (static vs periodic changes)	–	mean value, dominant frequencies	fast Fourier transform (FFT, FreMEn) (periodic changes)	–
Bescos *et al.* (2018) [18]	keyframe (graph)	bundle adjustment (3D, 6DoF)	x	x		x		camera (color, mono, stereo, RGBD)	pixel segmentation (Mask R-CNN), multi-view geometry changes (depth)	dynamic or movable classes (person, bicycle, car, motorcycle, airplane, bus, train, truck, boat, bird, cat, dog, horse, sheep, cow, elephant, bear, zebra and giraffe)	–	–	background inpainting
Sun *et al.* (2018) [130]	grid (occupancy, 3D)	point cloud alignment (3D, 6DoF)		x		x		laser (3D)	recurrent neural network (LSTM RNN) for cell semantic state transition	–	–	–	–

Table 2.4 (Continued)

DDE:	**1:**	**2:**	**3:**					**4:**	**5–9: Dynamics**				
Ref.	**Mapping**	**Localization**	**Indoor**	**Outdoor**	**Air**	**Ground**	**Water**	**Sensors**	**Detection**	**Labelling**	**Statistics**	**Prediction**	**Removal**
Lázaro *et al.* (2018) [22]	submap (2D point cloud, graph, 3DoF edges; g^2o)	point cloud alignment (2D, 3DoF)	x			x		wheel odometry, laser (2D)	ray tracing, scan-to-scan matching	–	–	–	cloud merging, outdated nodes removal
Boniardi *et al.* (2019) [23]	pose graph (graph, 2D point clouds, 3DoF edges), CAD floor plans	point cloud alignment (2D, 3DoF)	x			x		laser (2D)	scan-to-map matching	–	–	–	outdated nodes removal
Wang *et al.* (2019) [19]	keyframe (graph)	bundle adjustment (3D, 6DoF)	x	x		x		camera (RGBD)	semantic segmentation (FCIS), feature motion consistency check	static background, moveable class (person, vehicles, cup, chair, etc.)	–	–	moveable points removal
Zhang *et al.* (2019) [11]	grid (signed distance field, 2D)	point cloud alignment (2D, 3DoF)	x			x		laser (2D)	ray tracing (Bresenham algorithm), scan-to-map matching	static + semi-static + dynamic objects	–	–	dynamic objects removal
Ganti and Waslander (2019) [12]	keyframe (graph)	bundle adjustment (3D, 6DoF)		x		x		camera (stereo)	semantic segmentation (Bayesian SegNet)	static (road, sidewalk, building, wall/fence, pole, traffic light, traffic sign, vegetation, terrain) + dynamic (sky, person/rider, car, truck/bus, motorcycle/bicycle, void) classes	–	–	–
Song *et al.* (2019) [45]	keyframe (graph)	odometry (3D, 6DoF)		x		x		camera, IMU	pixel-wise motion attribute segmentation (unstable vs moving vs static, MD-Net)	unstable (sky, vegetation, terrain) + moving (human, vehicle, traffic light) + static (ground, flat, building, wall, fence, guard rail, bridge, tunnel, pole, pole group, traffic sign)	–	–	–
Pan *et al.* (2019) [41]	feature (point clusters)	odometry, reprojection minimization (3D, 6DoF)		x		x		camera (mono), laser (3D)	point clusters registration	–	clusters occurrence	–	dynamic point clusters removal
Wang *et al.* (2020) [50]	grid (occupancy, 2D)	particle filter (2D, 3DoF)	x			x		wheel odometry, laser (2D)	time series modelling	–	autoregressive moving average (ARMA)	autoregressive moving average (ARMA; static + periodic + aperiodic changes)	–
Yang *et al.* (2020) [42]	keyframe (graph)	visual odometry (3D, 6DoF)	x			x		camera (RGBD)	object detection (YOLOv3), geometry constraints	dynamic class (people)	–	–	–
Ding *et al.* (2020) [38]	point cloud (3D)	bundle adjustment (3D, 6DoF)		x		x		camera (color, stereo), laser (3D)	multi-session consistency, point clustering	–	observations count	–	intra (RANSAC, ray tracing) + inter-session (map matching) dynamic points removal
Yue *et al.* (2020) [43]	point cloud (3D), grid (occupancy, 3D)	–		x		x		camera (color, mono, thermal), laser (3D)	object detection (YOLOv3), 3D point cloud projection	dynamic class (people)	–	–	–
Singh *et al.* (2021) [46]	topological (BoW)	feature matching (location)		x		x		camera (stereo, RGBD)	pixel segmentation	dynamic class (sky, person, car, etc.)	–	–	down-weight dynamic class
Zhu *et al.* (2021) [13]	grid (occupancy, 2D)	particle filter (2D, 3DoF)	x			x		wheel odometry, camera, laser (3D)	object detection (YOLOv3)	static + semi-dynamic (chairs, parked cars) + dynamic (moving people / cars) objects, static + semi-dynamic maps	–	–	down-weight semi-dynamic objects removal
Guo and Liu (2021) [14]	keyframe (graph)	feature matching (3D, 6DoF)	x			x		camera (RGBD)	pixel segmentation (Mask R-CNN, depth region growing refinement), multi-view geometry segmentation (depth)	dynamic vs static / ground class	–	–	background inpainting

Table 2.4 (Continued)

DDE:	1:	2:	3:					4:	5–9: Dynamics				
Ref.	**Mapping**	**Localization**	**Indoor**	**Outdoor**	**Air**	**Ground**	**Water**	**Sensors**	**Detection**	**Labelling**	**Statistics**	**Prediction**	**Removal**
Thomas *et al.* (2022) [39]	sparse grid (hashmap, occupancy, 2D)	point cloud alignment (3D, 6DoF)	x			x		wheel odometry, laser (3D)	ray tracing (NN training), feature semantic labelling (KPConv)	feature semantic labelling (ground, permanent, movable, dynamic)	–	dynamic motion label (KPConv)	movable (localization) + dynamic (localization, mapping) points removal
Du *et al.* (2022) [32]	keyframe (graph)	reprojection minimization (3D, 6DoF)	x			x		camera (RGBD)	long-term consistent conditional random field (LC-CRF)	–	observations count, average reprojection error, symmetric epipolar distance	–	dynamic features removal
Xing *et al.* (2022) [8]	keyframe (graph)	feature matching (3D, 6DoF)	x	x	x	x		camera (gray, stereo), IMU	object detection (MobileNet V2 SSD), feature motion consistency check	short + long-term dynamics (STD, LTD)	–	–	–
Chen *et al.* (2022) [44]	point cloud (3D)	odometry (3D, 6DoF)		x		x		laser (3D)	object detection (PointPillars)	dynamic points (vehicles, pedestrians, cyclists, etc.)	–	–	dynamic objects removal
Kim and Kim (2022) [24]	keyframe (graph; iSAM2, GTSAM)	point cloud alignment (3D, 6DoF)		x		x		laser (3D), IMU	scan-to-map matching, multi-session consistency	high + low dynamic (HD, LD), positive (new points) + negative (disappeared) difference (PD, ND)	–	–	positive + negative difference maps (PD, ND)
Wu *et al.* (2022) [30]	keyframe (graph)	feature matching (3D, 6DoF)	x			x		camera (RGBD)	reprojection error (Student's t-distribution modelling), conditional random field (CRF) minimization	–	static score (Student's t-distribution	–	dynamic features removal
Fu *et al.* (2022) [131]	feature (plane, SDF)	–	x			x		camera (RGBD)	planes 2D height map comparison, planes SDF 3D validation	2D ternary change mask (changed, unchanged, unexplored)	–	–	–
Liu *et al.* (2022) [31]	keyframe (graph)	bundle adjustment (3D, 6DoF)	x			x		camera (RGBD)	motion consistency, reprojection error	–	static probability (reprojection error)	–	dynamic features removal
Deng *et al.* (2023) [15]	keyframe (graph)	bundle adjustment (3D, 6DoF)	x			x		camera (stereo, RGBD)	scan-to-map matching	dynamic + semi-static (periodic, abrupt changes) + static points	observations count, survival time, point persistence	Bayesian persistence filter (persistence probability), time series modeling (semi-static points)	dynamic points removal
Nielsen and Hendeby (2023) [132]	feature (multi-hypotheses tree)	–		x		x		wheel odometry, laser (2D)	multi-hypotheses testing	–	negative information (missed detections) + new landmark observations + false measurements + landmark entering FOV	–	hypotheses pruning / merging
Soares *et al.* (2023) [20]	keyframe (graph)	bundle adjustment (3D, 6DoF)	x			x		camera (RGBD)	object detection (YOLOv4), feature repopulation (back vs foreground)	dynamic (person) + movable (chair) + static objects	–	–	covariance static < movable < dynamic measurements
Stefanini *et al.* (2023) [25]	grid (occupancy, 2D)	–	x			x		wheel odometry, laser (2D)	scan-to-map matching, ray tracing	–	cells changed flag count	–	–
Yu *et al.* (2023) [47]	keyframe (graph)	bundle adjustment (3D, 6DoF)	x			x		camera (RGBD)	semantic segmentation (lightweight Mask R-CNN + MobileNetV2), motion probability model	dynamic class (humans, animals, cars)	motion probability model (semantic mask + optical flow)	–	dynamic features removal
Hroob *et al.* (2024) [26]	grid (occupancy, 3D), point cloud (3D)	point cloud alignment (3D, 6DoF)		x		x		laser (3D), IMU	scan-to-map matching, long-term stability network (LTS-NET, PointNet++-based)	stable vs unstable points	point spatiotemporal stability score (LTS-NET)	point stability score (unsupervised learning, based on recorded sessions)	stability score threshold removal (localization)
Hroob *et al.* (2024) [27]	grid (occupancy, 3D), point cloud (3D)	point cloud alignment (3D, 6DoF)		x		x		laser (3D)	scan-to-map matching, stability score inference (modified MinkUNet14), scan-map discrepancy segmentation	stable vs unstable points	point spatiotemporal stability score (modified MinkUNet14)	point stability score (unsupervised learning, based on recorded sessions)	stability score threshold removal (localization)

Table 2.4 (Continued)

DDE:	1:	2:	3:					4:	5–9: Dynamics				
Ref.	**Mapping**	**Localization**	**Indoor**	**Outdoor**	**Air**	**Ground**	**Water**	**Sensors**	**Detection**	**Labelling**	**Statistics**	**Prediction**	**Removal**
Wang *et al.* (2024) [16]	grid (occupancy, 2D)	particle filter (2D, 3DoF)	x			x		wheel odometry, laser (2D)	object clustering (k-means), time series modelling	semi-static vs dynamic map	autoregressive moving average (ARMA), dynamic object density	autoregressive moving average (ARMA; static + periodic + aperiodic changes)	–
Yang *et al.* (2024) [29]	point cloud (3D)	–		x		x		laser (3D)	multi-scan consistency (OctoMap), multi-session consistency (k-NN spatial filter)	–	–	–	positive + negative difference maps (PD, ND)
Cheng *et al.* (2025) [34]	3D Gaussian splatting	gradient-based optimization (rendered RGB + depth + semantic maps)		x		x		camera (RGBD), laser (3D)	object segmentation (SegmentAnything), motion + temporal consistency	entering + vanishing + full observation	–	–	3D Gaussian-based encoding (dynamic objects)
Hu *et al.* (2025) [35]	object map (3D point cloud, color histogram, semantic category)	feature matching (3D, 6DoF)	x			x		camera (RGBD)	object detection (YOLOX), motion consistency, foreground vs background clustering (DBSCAN)	–	object static probability	–	dynamic points weighting (pose optimization)
Jia *et al.* (2025) [36]	3D Gaussian splatting	feature matching (3D, 6DoF)	x			x		camera (RGBD)	semantic segmentation (FastInst), motion consistency	–	–	–	dynamic objects removal (3DGS model)
Li *et al.* (2025) [17]	grid (hashmap, dynamic i-Octree)	point cloud alignment (3D, 6DoF)		x		x		laser (3D), IMU	scan-to-map matching	prior (multi-session) + temporary (short-term) + static map points	–	–	–
Wang *et al.* (2025) [33]	feature (point, line, plane)	bundle adjustment (3D, 6DoF)	x			x		camera (RGBD), IMU	semantic segmentation (DeepLab v3), motion consistency (optical flow), structural regularity check (Manhattan world)	–	dynamic probability (dynamic filter)	–	dynamic features removal (dynamic probability threshold)
Zhu *et al.* (2025) [28]	sparse grid (hashmap, 2D+3D, i-Octree)	odometry (3D, 6DoF)		x		x		laser (3D), IMU	scan-to-map matching, object clustering (region growing)	–	–	–	dynamic objects removal (scan filtering), ghost trails retroactive removal (i-Octree seach)

2.3.1 Map representation

Inspired by the human memory, Dayoub *et al.* [6] and Bacca *et al.* [7] adapt the multi-store model of Atkinson and Shiffrin [133] for mapping. This model divides memory into three stores: Sensory Memory (SM), which stores the most recent information from the sensors; Short-Term Memory (STM); and Long-Term Memory (LTM). Three mechanisms move information between memories: selective attention from SM to STM, rehearsal from STM to LTM, and the retrieval mechanism to move unused information from LTM back to STM. Dayoub *et al.* [6] implement two types of state machines for rehearsal and retrieval mechanisms of the STM and LTM. In rehearsal, an STM feature moves closer to LTM or moves back to the initial state (or is forgotten if already in that state), depending on whether it is observed consecutively or not. Similar for retrieval, where a feature in LTM moves to the initial state or closer to being forgotten if it is not observed in the current view. Consequently, LTM and STM retain the most static and dynamic features, respectively, based on their observability in the current view. In a changing environment, the method reduced localization failure rates compared to a static view. Instead of using a state machine, Bacca *et al.* [7] implement a Feature Stability Histogram (FSH) depending on the feature observability to distinguish between STM and LTM features using k-means clustering. This modification allows an input feature in SM to bypass STM and become part of LTM, depending on the feature strength. The method was able to filter out pedestrians from 2D laser and camera data and achieved a more accurate representation of the environment compared to a static approach.

Although Biber and Duckett [4] do not adopt specific memory mechanisms, they implement STM and LTM maps. The method implements a dynamic map as a set of local maps, each containing submaps that represent different timescales. The timescale parameter for each submap determines when to add samples from 2D laser scans probabilistically. The dynamics of the environment are represented by using 5 different timescales, where the smallest one ($\sim$3.1 s) represents an STM map updated at every instant, and the other 4 timescales ($\sim$0.43 runs, 0.43 days, 3.1 days, and 13.5 days) are LTM submaps updated after each session or daily. Instead of focusing only on the LTM maps as in Dayoub *et al.* [6] and Bacca *et al.* [7], Biber and Duckett [4] select the best representation of both STM and LTM maps that best explains the sensor data. In a 5-week experiment, localization improved with a dynamic map, in contrast to the degradation observed when only considering a static representation of the environment. Indeed, the usage of timescales led to static parts emerging as walls in LTM and dynamic elements disappearing from STM maps.

Similar to the works [6] and [7], two maps can provide a more stable and dynamic representation of the environment. In Walcott-Bryant *et al.* [10], an active map represents the most current state of the environment, including parts that did not change from previous passes and objects added to the environment. A dynamic map only saves the points of a 2D laser scan that changed over time. Wang *et al.* [19] use a tracking map with short-term static points and a long-term map containing only long-term static points, identified by a semantic segmentation module, with ORB-SLAM2 [134]. Also, Zhu *et al.* [13] create offline a semi-dynamic map and a static one, where the former has semi-dynamic objects (parked cars in a parking lot environment), and the latter has both static and semi-dynamic objects. The main goal of representing two different dynamics is to favor the more stable one in the long term. Walcott-Bryant *et al.* [10] and Wang *et al.* [19] use only static parts of the current environment representation (active and long-term maps, respectively) for localization. In the experiments, the approach [10] showed that it could identify static parts, though it was affected by false positives and negatives and by the blur effect in the 2D grid map. The method [19] improved the Absolute Trajectory Error (ATE) in a dynamic environment over ORB-SLAM2 [134] and DynaSLAM [18]. As for Zhu *et al.* [13], their Monte Carlo Localization (MCL) framework (equivalent to particle-filter-based localization) reduces the weight assigned to observations of moved semi-dynamic objects. The method improved the localization in a parking lot compared to a standard MCL.

As for a multi-hypothesis representation, Nielsen and Hendeby [132] propose a framework to track landmark positions using Markov chains to model transitions between positions. The method [132] groups position hypotheses by landmark signature, and selects, according to incoming observations, the position within each group with the highest probability as the active one. The hypothesis likelihood depends on the match between the landmark measurements and the landmark in the map, as well as on the number of missed detections, *i.e.*, landmarks with a probability of being seen in the current pose but not represented in the observations. Although the work [132] maintained an up-to-date representation of the actual landmark positions in simulated and real experiments, the proposed approach always assume uniquely identifiable landmarks.

Li *et al.* [17] leverage a map representation based on voxelized maps and octree structures to handle environment dynamics. The work [17] organizes prior, temporary, and static map points within hashed voxel blocks, each containing a Dynamic i-Octree [135] for efficient local map loading, radius-neighbor search, and incremental map updates. The octree representation is not only a storage/indexing structure but also a mechanism that allows the system to adapt when the robot moves between mapped and unmapped areas or when the environment changes. This mechanism promotes temporary observations into the static map only when sufficient evidence accumulates. Experiments demonstrated the effectiveness of the method, in which a prior map was dynamically updated to reflect changes caused by parked vehicles and growing trees, while minimizing the inclusion of dynamic objects, such as passing vehicles and pedestrians.

2.3.2 Map matching

Another trend in the literature is the identification of the scene dynamics by comparing the current observation to the map. Assuming a prior vector map as a permanent map, Biswas and Veloso [9] determine the probability of long-term observed features by the 2D laser scan-to-map matching distance. Short-term features are identified by the scan-to-scan matching distance, while the remaining features are considered dynamic and not used for localization. Compared to MCL with a static map and to Tipaldi *et al.* [48], Biswas and Veloso [9] obtained lower localization error in a parking-lot environment. However, the method would not handle semi-static changes. Instead of using a permanent map, Zhang *et al.* [11] maintain an Signed Distance Field (SDF) representation based on a prior occupancy map. The proposed approach rejects dynamic points identified by range flow and updates the SDF-based map with semi-static changes observed in the scan-to-map difference. Compared to MCL in a semi-static environment, Zhang *et al.* [11] had lower pose errors and an improved representation of the environment.

Similarly, Fu *et al.* [131] use SDFs in mapping, proposing the PlaneSDF representation, in which a scene is a set of planes, each with an associated SDF volume. This volume describes geometric details of the objects attached to their associated plane. Then, a change detection algorithm compares 2D height maps and performs 3D geometric validation between two PlaneSDF-based maps to compute a change mask. However, the method [131] failed to detect changes in ambiguous situations, such as object occlusions. Boniardi *et al.* [23] detect semi-static changes by leveraging ICP scan-to-map consistency and a Computer-Aided Design (CAD) model of the environment, and update the map accordingly. The proposed approach [23] was capable of maintaining a consistent map even with substantial environment reconfiguration. Du *et al.* [32] minimize the Gibbs energy defined on the proposed Long-term Consistent Conditional Random Field (LC-CRF) to detect dynamic points, assuming that these points often exhibit large reprojection errors in frame-to-map matching and that points tend to share dynamic properties with their neighbors. In a dynamic scene, LC-CRF [32] achieved lower ATE than ORB-SLAM [136]. Unlike LC-CRF [32], the Dynamic visual Feature Removal SLAM (DFR-SLAM) proposed by Liu *et al.* [31] does not look for similar dynamic properties in the features' neighbors. Instead, the method [31] assumes that static features exhibit consistent motion over a long time and that dynamic features exhibit distinct motion patterns, labeling them using the proposed motion-consensus filtering algorithm. DFR-SLAM [31] formulates a binary classification problem using a graph-cut optimization framework, assigning penalties to connected visual features across instances that do not share the same label. The method [31] had a lower ATE than ORB-SLAM2 [134] in dynamic scenarios.

Furthermore, Pan *et al.* [41] and Ding *et al.* [38] leverage clustering properties of the observations, evaluating the observation count when matching the scan with the map. The proposed approach segments the points of a 3D LiDAR point cloud into clusters, assuming that dynamic points do not appear frequently at the same location. The map only considers clusters that appear in the same location more than 10 times. As for Ding *et al.* [38], the method builds on the assumption that the dynamic and static parts of the environment exhibit clustering properties relative to their neighbors (similar to Du *et al.* [32]). The number of observations across multiple mapping sessions, combined with their consistency relative to neighbors, determines whether a map point is static throughout the sessions. Both representations of the environment in Pan *et al.* [41] and Ding *et al.* [38] were stable to structural changes in the environment.

Building on multi-session consistency, Kim and Kim [24] classify dynamic points into high- and low-dynamic categories. An initial processing step removes high-dynamic range points using a range-image-based discrepancy. This step is followed by a multi-session consistency check in which aligned sessions are compared via kd-tree searches to identify low-dynamic points. By parsing negative (removed data between consecutive sessions) and positive differences (new data appearing in the environment between sessions) instead of saving the whole mapping sessions, their representation achieves a 60% reduction in memory usage compared to maintaining full session histories. Similarly, Yang *et al.* [29] utilize a clean initial base map and identify changes in incoming sessions using a k-Nearest Neighbor (NN)-based spatial filter. The proposed approach segments points into base-difference, coexist, and session-difference categories, while implementing a map version-control system based on positive and negative differences to maintain the environment representation throughout time. To accelerate this process, Yang *et al.* [29] employ 2D Bird-Eye-View (BEV) descriptors to pinpoint intensity differences in 3D data. A similar memory reduction compared to Kim and Kim [24] was observed in the experiments, increasing memory efficiency by 40.1%–94.2% across datasets, while retaining at least 5–7% more static points compared to [24].

The concept of ray tracing is also used by the included works to handle dynamic changes. Lázaro *et al.* [22] use ray tracing to exploit free space information. When comparing two 2D point clouds from a viewpoint, ray tracing identifies new objects added to the scene (observed points closer to the viewpoint than the old ones) and outdated information (observed points farther away), allowing dynamic change detection and providing an up-to-date representation for localization. Also, Lázaro *et al.* [22] merge points on the same ray that are close to each other into a new point based on the weighted average of their covariances. Experiments showed improvements in map definition and removal of non-static obstacles. Stefanini *et al.* [25] adopt a 1-to-N ray-casting strategy to improve map merging in the face of environment dynamics. Rather than a standard point-to-point comparison, the method [25] compares measurement points against a set of expected hit points along virtual neighboring range measurements to detect scene changes more robustly.

Given that 3D ray tracing is memory-intensive and requires dense map representations, Pomerleau *et al.* [37] use the sparse point cloud from a 3D laser directly. The map points are associated with every single reading via small conical apertures in spherical coordinates, updating the observed points closer than the mapped ones, and leaving the farther ones untouched. The approximation of ray tracing results is used to update the probability that points in the map are dynamic, using a Bayesian approach. The probability of being dynamic can be used in ICP to avoid trusting dynamic points. In the experiments, the method [37] produced a more precise and cleaner map than standard ICP matching. Instead of weighting the map points, An *et al.* [129] propose the Dynamic Edge Link (DEL) to model the dynamics of a pose graph's edges rather than the data itself. The observation of moving obstacles between two poses gradually decreases the weight of the respective edge until it no longer detects the obstacle. Integrating DEL in an exploration scheme, nodes with an edge weight average lower than a certain threshold, meaning frequent moving obstacles or changed structure near that node, are not considered for exploration because the robot may be unable to move to that position.

Although the standard NDT representation does not model free space, the works of Einhorn and Gross [40, 128] and Saarinen *et al.* [21] use NDT with occupancy maps to explicitly model free space and adopt an exponential weighted moving average and covariance for new measurements, giving them stronger influence than old ones. Einhorn and Gross [128] propose a generic 2D/3D mapping using NDT and occupancy maps. The hit cells from scan-to-map alignment are updated incrementally with exponential weighting, based on the current observation. The other cells along the sensor beam, potentially empty, are updated using the standard update rule for occupancy maps, based on the log-odds of the occupancy value and the inverse range sensor model. Instead of using the standard occupancy map update, the sensor model in Saarinen *et al.* [21] relies on the inconsistency between observations and the map. Also, the occupancy value describes the confidence of the NDT based on past observations. As for Einhorn and Gross [40], the proposed approach defines two occupancy map probabilities: occupancy and statically occupied. The former is updated based on the sensor model (2D/3D generic beam sensor) and scan-to-map alignment, and the latter slowly adapts to high-probability static objects in the environment. The static occupancy probability follows the proposed ad-hoc model, which is parameterized to control how quickly the static occupancy probabilities are adapted, depending also on the occupancy probability itself. The works [128] and [40] handled both semi-static and dynamic changes, maintaining a consistent, up-to-date representation of the environment. In contrast, the method [21] favored long-term static structures in dynamic environments.

2.3.3 Dynamic objects detection

In order to detect dynamic objects, Yue *et al.* [43] propose a collaborative dynamic-mapping approach for human detection using visual and thermal images and a 3D LiDAR. The YOLOv3 object detection algorithm extracts the bounding boxes from the images relative to humans. Projecting the 3D point cloud onto images creates a static point cloud for localization and mapping of each robot, filtering out points corresponding to humans. In the experiments, removing dynamic objects produced a more accurate relative transformation of the collaborative maps than not removing them. Zhu *et al.* [13] also use YOLOv3 to extract bounding boxes of dynamic classes (parked cars) from an RGB image, and the projection of 3D LiDAR points allows the creation of the static and semi-dynamic maps required for localization. Even though the method [13] does not discard dynamic objects, the localization module reduces the weight of the corresponding observations.

Furthermore, Sun *et al.* [130] adapt the PointNet network for object recognition (pedestrian, cyclist, car, or background) to classify 3D LiDAR scan points. The proposed Recurrent-OctoMap maintains occupancy and semantic information, where the latter specifies the cell's semantic state and the prediction's probability. An LTM Recurrent Neural Network (RNN) learns the transition between cell states. In a long-term experiment, the method improved its 3D semantic map compared to a standard Bayes update. In contrast, Chen *et al.* [44] use PointPillars for dynamic object removal within a LiDAR SLAM system employing Lidar Odometry and Mapping (LOAM) [137]. Given the bounding boxes predicted by PointPillars, the proposed approach crops all points from the 3D point cloud within those bounding boxes for later use in localization and mapping. Comparing localization and mapping errors before and after introducing dynamic object removal, localization accuracy remained similar in the experiments, though the maps created were much cleaner and sharper. However, localization performance sometimes decreased with dynamic object removal due to feature loss caused by false detections and a possible lack of features after removal.

Pixel-wise semantic segmentation is another way to identify dynamic objects. In addition to the semantic-descriptor proposed by Singh *et al.* [46], which describes the spatial distribution of semantic entities in the scene, the method assigns lower weights to features detected in the sky and in dynamic classes (*e.g.*, person, car, *etc.*) from EdgeNet semantic segmentation. The proposed approach [46] achieved higher accuracy than SeqSLAM [138], FAB-MAP [139], and Bag-of-Words (BoW) [140]-based place recognition methods in a highly dynamic outdoor experiment. Similarly, Yu *et al.* [47] implement a lightweight pixel-wise semantic segmentation network using MobileNetV2 as a backbone and building upon the ORB-SLAM2 [134] framework, also focusing on dynamic object classes such as humans, animals, and cars. Instead of down-weighting dynamic classes, all dynamic visual features are removed from the image using semantic annotations to obtain a 3D static semantic map. Compared to base ORB-SLAM2 [134], localization accuracy improved significantly in dynamic scenes, with an average Root Mean Square Error (RMSE) reduction of 95%.

Instead of identifying object classes, Song *et al.* [45] propose the MD-Net Convolutional Neural Network (CNN) to segment a grayscale image into unstable, static, and moving pixel points, only using static points for localization. The localization error was reduced compared to not estimating the pixel dynamic attribute. Ganti and Waslander [12] propose the Semantically Informed Visual Odometry (SIVO) to improve the performance of ORB-SLAM2 [134] by using the Bayesian neural network SegNet for segmentation and for computing network uncertainty. SegNet is trained to distinguish different object classes for identifying dynamic objects (sky, car, truck/bus, person/rider, motorcycle/bicycle, and void) from static ones (road, traffic sign, building, wall/fence, pole, vegetation, sidewalk, traffic light, and terrain). The proposed method considers only static keypoints as input to ORB-SLAM2 [134], thereby reducing most of the state's uncertainty (including network uncertainty). SIVO removed uninformative and dynamic keypoints from the current frame. However, rejecting potential dynamic objects without verifying their motion reduced the module's localization performance in certain scenarios. Soares *et al.* [20] propose Changing-SLAM built on ORB-SLAM3 [141]. Each feature keypoint is initially classified as dynamic, movable, or static by the object detection thread (YOLOv4) based on semantic information, and two keypoints belonging to the same object cannot have different classifications. Then, the proposed method employs feature repopulation to correctly classify features within a bounding box that are not dynamic. Experiments validate the feature repopulation technique, which keeps keypoints belonging to the background but within the dynamic object class. Comparing the proposed method to DynaSLAM [18] and ORB-SLAM3 [141], even though DynaSLAM [18] had overall the best results, Changing-SLAM offered a good trade-off between accuracy and speed in dynamic environments.

Spatial consistency can improve the identification of moving objects. Bescos *et al.* [18] propose DynaSLAM, which uses multi-view geometry verification to improve the semantic labeling of potential dynamic classes generated by a Mask RNN. Although DynaSLAM [18] outperformed ORB-SLAM2 [134] in highly dynamic scenarios, its performance reduces in slower dynamics. Similarly, Wang *et al.* [19] also use geometric verification to improve the labeling of dynamic objects, which are obtained using a ResNet-based network. Verification is based on reprojection error. The method [19] improved the ATE over DynaSLAM [18] and ORB-SLAM2 [134] in a scenario with movable objects. Guo and Liu [14] also use a Mask RNN network for pixel-wise segmentation and object instance labels, similar to DynaSLAM [18]. Rather than using multi-view geometry verification solely to improve semantic labeling, it is also used to detect untrained moving objects. The approach [14] outperformed DynaSLAM [18] in terms of ATE in dynamic scenes.

Instead of using semantic segmentation, the Semantic and Geometric Contraints Visual SLAM (SGC-VSLAM) proposed by Yang *et al.* [42] uses YOLOv3 to extract bounding boxes for dynamic objects, thereby also improving ORB-SLAM2 [134]. A constraint based on epipolar geometry improves the labeling. SGC-VSLAM decreases the RMSE of the ATE by 96% compared to ORB-SLAM2 [134] in highly dynamic environments. However, like DynaSLAM [18], its performance decreased at lower dynamics. Also, Xing *et al.* [8] propose the DE-SLAM to deal with Short-Term Dynamics (STD) and Long-Term Dynamics (LTD) at the same time. A MobileNetV2 identifies bounding boxes for movable objects (cars, persons, etc.) and classifies them as STD. A motion check of STD elements recognizes all moving objects in the current keyframe. As for LTD, DE-SLAM uses Histogram of Oriented Gradients (HOG) features extracted from Oriented FAST and Rotated BRIEF (ORB) keypoints to improve its invariance to illumination changes. In the experiments, DE-SLAM improved the localization over ORB-SLAM2 [134] in a changing environment. All of these methods, which use geometric constraints to improve semantic labeling, rely solely on static features for localization and mapping.

Moreover, Wang *et al.* [33] formulate a dynamic filter on top of ORB-SLAM3 [141] that fuses information from semantic segmentation and structural constraints (a Manhattan-world assumption with structure constraints between planar features) to identify both semantically identifiable and unidentifiable dynamic objects. The authors use the DeepLab semantic segmentation algorithm, capable of identifying 20 objects, including people and chairs, and leverage optical flow to detect abnormal dynamic features. Then, the dynamic filter computes long-term dynamic object probabilities considering structure regularity (related to the distance error of feature points to the epipolar line constraint), Inertial Measurement Unit (IMU) measurements (reprojection error of feature points from IMU pose estimation), and forward/backward projection (reprojection error of co-visible features onto the current frame). In experiments, compared to ORB-SLAM3 [141], it achieved up to 92% average RMSE improvements. Omitting structural regularity from the proposed approach [33] led to a marginal reduction in localization accuracy, whereas excluding the dynamic filter module altogether resulted in notable trajectory deviation. Zhu *et al.* [28] introduce the Incremental Static-referenced Dynamic Object Removal (ISDOR) module to detect and suppress moving objects in real time from 3D LiDAR data. The module leverages static structures (*e.g.*, ground) to identify points lying above static regions, vertically suspended points with significant height, or points adjacent to previously flagged dynamic areas. Once a portion of a dynamic object is identified with high confidence, the complete object is recovered via region growing and clustering. In experiments, ISDOR segmented dynamic objects visually align with the ground-truth, accurately isolating dynamic points while preserving static structure, including removing ghost trails in the datasets.

Another common approach is to formulate static scores based on spatiotemporal consistency to identify and handle dynamic elements in the scene. Wu *et al.* [30] formulate a static score estimation of the landmark to obtain initial static-dynamic labelling, and a Conditional Random Field (CRF) is constructed to refine the labels. Based on ORB-SLAM2 [134], the static score is estimated from the statistical properties of the reprojection error and then propagated to subsequent frames. Similar to Du *et al.* [32], the authors consider that adjacent landmarks in 3D space or 2D images tend to have the same dynamic-static label. A CRF that considers neighbor motion and historical observations is built on a fully connected graph of all landmarks, improving spatiotemporal consistency of static-dynamic labels. Comparing the accuracy and efficiency to the base ORB-SLAM2 [134] and to Du *et al.* [32], improved ATE and Relative Pose Error (RPE) by 80–90% over ORB-SLAM2 [134]. Although Du *et al.* [32] reported the lowest RMSE for ATE and RPE, the proposed work [30] achieved similar accuracy with only a few centimeters of disadvantage, while taking 5 times less computing time for dynamic landmark removal than Du *et al.* [32]. Hu *et al.* [35] employ YOLOX for object detection on input images, while compensating for missed detections using an EKF (predict the state of the bounding box of a certain object already mapped) and the Hungarian algorithm (associate EKF-predicted bounding boxes with those detected in the current frame). Authors formulate a static probability distribution over keypoints based on motion consistency (via projection and epipolar constraints). This static probability is used as a weight in camera pose estimation and optimization. Experiments compare the proposed approach [35] with several state-of-the-art algorithms, including DynaSLAM [18] and ORB-SLAM2 [134]. The proposed approach [35] achieved competitive localization accuracy while offering a good balance between real-time performance and localization accuracy.

More recently, the literature has explored dense environment representations and methods for handling moving elements. Cheng *et al.* [34] render the environment through a 3D Gaussian Splatting (3DGS) model. The authors use the SegmentAnything model to generate a pixel-wise semantic image, allowing correspondence between the features and their respective object labels. The objects are also classified into entering (objects entering the frame), vanishing (objects leaving the frame), and full (objects fully visible within the frame). A weighting factor accounts for the temporal decay of feature relevance (older features having less influence). Semantic data from the SegmentAnything model is integrated and rendered using a 3DGS formula. Then, the method [34] uses Gaussian-based encoding to represent dynamic objects, considering semantic attributes, pose, and motion velocity information, to avoid generating noisy Gaussian points from dynamic objects that can affect model convergence. The proposed approach [34] accurately rendered dynamic objects in experiments. Similarly, Jia *et al.* [36] propose the DGS-SLAM to reconstruct a static, consistent, and dense representation of the surroundings by excluding dynamic objects from the 3DGS model. The proposed approach [36] identifies dynamic objects using motion consistency (a dynamic projection threshold to evaluate the motion state of feature points) and object segmentation, with FastInst instance segmentation to obtain pixel masks of objects. When generating new Gaussian points from keyframes, each point is assigned a label indicating whether it belongs to an object or a static background, enabling 3D-2D association. DGS-SLAM achieved the best pose estimation results across most experiments, including tracking, compared to other 3DGS-based and Neural Radiance Field (NeRF)-based SLAM systems.

2.3.4 Prediction modeling

In the included works, Markov processes are used to predict the dynamics of the environment. Tipaldi *et al.* [48] use a dynamic occupancy grid and exploit the stationary distribution and the state holding time associated with Hidden Markov Models (HMMs) on a 2D grid. The method [48] uses past observations from each run to learn the state transition probabilities iteratively and estimate the HMM parameters. Then, localization can infer how often a dynamic object is expected to be seen in the environment and for how long. Comparing the proposed HMM-based localization to MCL using a standard grid, the former had a lower localization failure rate than MCL. Indeed, work [48] could handle both high-dynamics (moving cars) and low-dynamics (parked cars). Rapp *et al.* [49] implement a semi-Markov process extended by a Levy process to model time dependence in the state-holding time of Markov processes, also predicting, as in Tipaldi *et al.* [48], the expected retention time for each cell in a specific state. In the experiments, the proposed model by Rapp *et al.* [49], integrated into MCL, improved the classic MCL in a dynamic environment.

Environment dynamics may exhibit periodic patterns. Assuming periodic change patterns, Krajník *et al.* [52] propose Frequency Map Enhancement (FreMEn) to model the probability of occupancy or feature visibility in a grid as a combination of harmonic functions associated with periodic processes. FreMEn [52] uses spectral analysis (Fourier transform) to compute the harmonic functions and predict the future state with a given confidence. In a changing environment, FreMEn [52] outperformed a static map in terms of localization error by selecting the most likely visible features at each location. Santos *et al.* [51] adopt FreMEn [52] within an exploration scheme, where the planner predicts which areas are more likely to change at a certain time and generates the subsequent locations to explore. The experimental results showed that accounting for environment dynamics increases the amount of information gathered compared to static models.

Unlike FreMEn [52], Wang *et al.* [50] model both aperiodic and periodic changes using an Auto-regressive Moving Average Model (ARMA) model [142]. This model describes time series as stationary stochastic processes in terms of polynomials. While FreMEn [52] can update its model recursively online, ARMA [50] is updated only once a day based on past observations. However, the model by Wang *et al.* [50] achieved higher prediction accuracy than FreMEn [52] and a lower localization failure rate than both FreMEn [52] and Tipaldi *et al.* [48]. Wang *et al.* [16] extend the original work of Wang *et al.* [50] by adding long-term path planning to generate feasible paths with low localization uncertainty and fewer dynamic objects, while introducing new spatial representation models (semi-static vs dynamic density maps) and improving parameter calculation. In the experiments, the authors simulate a large-scale parking lot with periodic and aperiodic changing car patterns. As already shown in Wang *et al.* [50], since FreMEn [52] is based on a spectral model, it cannot effectively handle non-periodic growth patterns compared to ARMA models [16, 50]. Also, when evaluated in a real parking lot, the system proposed by Wang *et al.* [16] was able to predict more accurately for most parking spaces than FreMEn [52], being more stable across various real-world change patterns.

Den *et al.* [15] build a time series for each map point to predict their future states. The proposed approach uses a Bayesian Persistance Filter (BPF) and survival analysis to estimate the likelihood that each map point will still exist at a future time. The system classifies the map points as static or semistatic based on its persistence probability, in which points with a persistence probability lower than a certain threshold are considered dynamic ones and removed from the map. In the experiments, the authors compared the proposed prediction algorithm [15] to FreMEn [52] and ARMA [142] models in terms of predictive capabilities. The proposed method outperformed other prediction methods. As for localization accuracy, BPF-based localization had lower ATE in low-dynamic environments compared to ORB-SLAM2 [134], ORB-SLAM3 [141], and DynaSLAM [18]. In high-dynamic scenes, although the proposed method improved accuracy over ORB-SLAM2 [134] and ORB-SLAM3 [141], DynaSLAM [18] had lower ATE metrics.

Instead of modeling the dynamics in the map, Thomas *et al.* [39] use a KPConv network to predict online dynamic motion labels for points given a single 3D LiDAR as input. The method [39] is a self-supervised learning approach with two main modules: PointMap and PointRay. PointMap is an ICP-based SLAM algorithm to provide a point cloud map for the annotation process. PointRay uses a similar approach to Pomerleau *et al.* [37] to approximate ray tracing using spherical coordinates for obtaining the training annotation of dynamic labels: permanent (static points over all sessions), ground (to avoid ray tracing ground samples), and long-term (still objects in single sessions but relocated between sessions) and short-term (dynamic objects) movables, with the localization not considering the latter two. In simulation experiments, PointMap with the proposed prediction module yielded lower localization pose errors than the standard MCL algorithm.

Moreover, Hroob *et al.* [26] propose the Long-Term Stability NETwork (LTS-NET). This regression network estimates a stability value for each point in a 3D LiDAR point cloud, thereby providing a spatiotemporal stability measure. LTS-NET is built on PointNet++ and uses labels generated unsupervised from previous observations and the map, with the NDT employed for scan matching. Once trained, the network is applied online to identify stable points in the current LiDAR session, filtering out unstable observations. Experiments showed that filtering scans based on long-term stable features significantly enhances localization performance and robustness in dynamic environments, enabling the network to identify and filter long-term stable objects. These experimental results led to improved localization performance compared to not using network inference on stability scores. As for Hroob *et al.* [27], the authors formulate a more advanced method that also relies on NDT for scan matching, but replaces PointNet++ with a modified MinkUNet14 to process sparse 4D tensors containing spatial coordinates and time data. Rather than inferring stability primarily from training data, this method explicitly models the spatio-temporal discrepancy between the current LiDAR scan and the point cloud map. Scan and map tensors are merged into a unified sparse 4D representation, after which Hroob *et al.* [27] prune voxels containing only map timestamps, since the goal is to infer stability scores only for scan points. By directly exploiting scan-map discrepancies, the method [27] achieved stronger generalization in the experiments. Indeed, it outperformed LTS-NET [26] in segmenting unstable elements such as humans, vegetation, parked cars, and pedestrians, cases in which LTS-NET [26] was less effective, likely due to its stronger dependence on object shapes observed during training.

2.4 Summary and Conclusions

SLAM is one of the most fundamental problems in robotics. When a robot lacks prior knowledge of the environment and does not know its own pose, SLAM enables an AMR to build a map of its surroundings while simultaneously localizing itself within it. A basic strategy for solving the SLAM problem is to continuously align the latest sensor observation with the map up to the current instant, then integrate the new measurement into the map using the estimated alignment pose, as in scan-matching systems such as HectorSLAM [72]. However, a pure scan-matching approach does not explicitly address loop closure, which limits the correction of accumulated drift when the robot revisits a previously mapped place. Another popular family of approaches is based on particle filters, specifically Rao-Blackwellized Particle Filter (RBPF), as in GMapping [74] (adaptation of RBPF to 2D occupancy grid mapping). In these methods, the posterior distribution over part of the state, such as the robot's trajectory, is represented using particles. At the same time, the remaining variables are modeled using Gaussians or other parametric distributions, resulting in higher computational efficiency than standard particle filters [78]. More recently, pose-graph formulations have become a dominant paradigm for solving the SLAM problem. In graph-based SLAM, the front-end identifies constraints, such as odometric relations between consecutive poses, loop closures with previously visited poses, and landmark observations from one or multiple viewpoints, by establishing correspondences between new observations and the current map. The back-end then refines the graph by jointly minimizing the inconsistency of all constraints [87]. An implementation of this formulation for 2D laser scanners and wheel odometry is SLAM Toolbox [95]. Because the pose-graph formulation retains access to the full set of measurements and constraints during map optimization, it allows the use of improved linearization, loop-closing, and data-association strategies, which is one of the main reasons why graph-based SLAM often outperforms purely filter-based approaches. Nevertheless, pose-graph methods usually exhibit at least linear growth of the graph with the explored area, which makes graph sparsification, submapping, and map-maintenance strategies important for long-term operation [65, 78].

In a static environment, the robot would only need to map it once, since the map and incoming sensor data would remain consistent over time. Nonetheless, real environments contain moving agents, temporary occlusions, furniture rearrangement, seasonal or illumination changes, and other sources of temporal variability. Long-term SLAM addresses precisely these issues. In our previous work [65], we conduct a systematic literature review on long-term localization and mapping following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) [98] guidelines. To the best of our knowledge, before that publication, no other work had reviewed the broader topic of long-term SLAM in a unified way, covering appearance variation for localization and place recognition, the modeling and handling of scene dynamics, redundant data removal from maps, multi-session mapping, computational aspects of long-term operation, datasets, and evaluation metrics. Using a systematic methodology to ensure repeatability and reproducibility, we review 142 works across 6 scientific databases. Overall, the scientific community have shown increasing interest in long-term localization and mapping. The subtopic of SLAM in dynamic environments, which is the focus of this PhD Thesis, has also experienced clear growth over time.

Subsequently, this PhD Thesis builds on the results of our systematic review, narrowing the discussion to long-term SLAM in dynamic environments and updating the search date of the last full inquiry to February 13, 2026. This update results in 53 works analyzed in Section 2.3. A trend in the literature is that most methods still rely on conventional SLAM map representations, such as graphs, occupancy grids, point clouds, or feature maps, and then augment them with mechanisms to reason about dynamics. In other words, the map remains primarily a static structural representation, while dynamic behavior is encoded as an additional property of cells, points, features, or objects. A recurring strategy, especially in human memory-inspired works, is to organize information into Sensory Memory (SM), Short-Term Memory (STM), and/or Long-Term Memory (LTM) [4, 6–9]. These works use selective attention, rehearsal, retrieval, and pruning policies to determine which observations should be included in the persistent map and which should remain temporary or ignored. Another common trend is to partition the map into static, semi-static, and/or dynamic components [10–17], or equivalently into permanent, movable, and dynamic layers [18–20]. In these formulations, static or permanent structures are expected to remain stable over long periods; semi-static or movable elements are objects that may persist for some time but can change between visits; and dynamic elements are objects currently moving or highly unstable in the sensor field of view.

Furthermore, methods that exploit data consistency are common in the literature for identifying scene dynamics. One typical strategy is to align the current sensor observation with the current map or with previous observations and then detect mismatches. Depending on the sensing modality, these spatial inconsistencies manifest as scan-to-map or scan-to-scan mismatches with range sensors [6, 10, 11, 15, 17, 21–29], reprojection errors with visual modalities [19, 30–33], motion inconsistencies in tracked features [8, 19, 31, 33–36], ray-tracing discrepancies that reveal free-space discrepancies or newly added objects [4, 9, 11, 21, 22, 25, 37–39], and multi-session inconsistencies across repeated traversals of the same place [24, 29, 38]. Clustering is often used as a complementary step to group nearby inconsistent observations and turn isolated outliers into more coherent dynamic hypotheses [7, 16, 28, 35, 38, 40, 41]. Also, semantic perception has become a central component of long-term SLAM in dynamic environments, particularly in vision-based approaches. Object detection and semantic segmentation are frequently used to identify categories that are likely to be dynamic or movable, such as people, vehicles, bicycles, or animals [8, 13, 20, 35, 42–44]. Pixel-wise segmentation is also employed in the literature discriminate between dynamic and static image regions and avoid excessive feature depletion compared to bounding-box semantic segmentation [12, 14, 18, 19, 33, 34, 36, 45–47]. Indeed, bounding-box detectors often require additional geometric constraints, background/foreground refinement, or feature repopulation mechanisms to avoid incorrectly rejecting useful static features within coarse detections [20].

Statistics are widely used in the literature to describe and, in some cases, predict scene dynamics. Early methods often relied on relatively simple quantities, such as hit-and-miss counts [6], change scores [10], observation counts [15, 32, 38], or feature stability histograms [7], to estimate how often a cell, point, or feature was observed consistently over time. Later works introduced richer temporal models, including HMM [48], semi-Markov processes [49], ARMA models [16, 50], persistence filters [15], and frequency-analytical models based on the Fast Fourier Transform (FFT) [51, 52]. These models are typically not used to forecast detailed future trajectories of dynamic objects, but rather to estimate a persistence, reliability, or staticness score for map elements, or even to identify periodic temporal patterns. Possible uses of predicting dynamics include inferring whether a region of the environment is stable, periodically occupied, or likely to change. Such information can then support more robust localization systems, more selective map updates, and even planning strategies that avoid highly dynamic areas or unfavorable times for revisiting them.

Most methods in the literature aim to protect localization from transient or unstable observations by relying primarily on static maps, or at most on static plus semi-static components, while excluding clearly dynamic elements from the pose-estimation process. In some approaches, dynamic and semi-static observations are removed before being inserted into the long-term map [9–11, 15, 22, 23, 26–28, 30–33, 36, 38, 39, 41, 44, 47, 129]. In contrast, other approaches do not discard those observations. Instead, dynamic observations and possibly semi-static ones are down-weighted during scan or feature matching, or are assigned larger covariance to reduce their influence on the pose estimate [13, 20, 37, 46]. Vision-based methods often use masking and background inpainting to remove dynamic image regions while preserving a usable visual observation for tracking and mapping [14, 18]. Overall, the dominant trend is that long-term SLAM systems in dynamic environments remain fundamentally static-map-oriented. Rather than building a fully dynamic world model, they estimate the stability of observations and decide whether to keep them, temporarily down-weighted, or remove them altogether. This trend makes the notion of persistence, rather than motion alone, one of the central concepts in the current long-term SLAM literature for dynamic environments.

In conclusion, the literature on long-term SLAM in dynamic environments shows progress in modeling persistence, detecting dynamic elements, and improving localization robustness. Still, the literature also reveals several gaps that motivate this PhD Thesis. Most existing approaches remain fundamentally static-map-oriented: they typically identify, mask, down-weight, or remove dynamic observations. However, current methods rarely maintain a continuously updated map representation that explicitly replaces outdated structure with the most recent consistent observations. In addition, the literature tends to treat mapping quality, localization robustness, and long-term map maintenance as partially separate problems, with limited integration between online dynamic-aware operation and offline global refinement. Another issue is scalability. Pose-graph and map representations often grow over time, even when the robot revisits known areas, introducing redundant data, increasing computational cost, and degrading map quality through accumulated noise and distortion. To address these limitations, this PhD Thesis proposes a unified SLAM framework for ground mobile robots in dynamic indoor environments that combines: online ray tracing-based map updating to remove outdated or redundant observations; relocalization on existing graph structures to prevent unnecessary graph expansion; offline generation of cleaner static maps by removing dynamic points; and offline global refinement to correct accumulated distortions. In parallel, this PhD Thesis researches geometric features, distance-map-based registration with robust kernels, and continuously updated local maps. The final goal is to actively incorporated environment dynamics instead of just filtering them out, to provide a more stable, scalable, and accurate SLAM solution in highly dynamic environments.

Chapter 3

Datasets

TBC

3.1 Mobile Platforms

TBC

3.1.1 5dpo three-wheeled omnidirectional robot

3.1.2 Modified Hangfa Discovery Q2 four-wheeled omnidirectional platform

3.1.3 IDEPA's tricycle autonomous robot

3.1.4 Avantgarde tricycle autonomous forklift

3.2 Faculdade de Engenharia da Universidade do Porto

TBC

3.3 IILABS 3D – iiLab Indoor LiDAR-based SLAM

TBC

IDEPA

Cappero

Avantgarde at COLEP

Kuka

3.4 Summary and Conclusions

TBC

Chapter 4

Line Fitting-Based Corner-Like Detector for 2D Laser Scanners Data

This chapter presents a novel line-fitting-based corner-like detector for 2D laser-scanning data [66], developed within the scope of this PhD Thesis (see Appendix C). Even though the most common way to represent 2D environments is via occupancy grid maps, an alternative is to represent the scene as a set of features. Accordingly, this chapter explores the extraction of features that may resemble geometric corners in the environment. Our methodology explores local line fitting at each scan point and the intersection of the fitted lines before and after the point, with respect to the angular scanning direction. Experimental results reveal viewpoint and geometric invariance of the corner-like features extracted using the proposed methodology, demonstrating the potential of this work for localization and mapping problems on mobile robots.

Thus, this chapter is divided into four sections. First, Section 4.1 introduces feature extraction algorithms for 2D laser scanner data from the literature. Section 4.2 outlines our methodology for extracting corner-like features from 2D laser data using local line fitting and multiple neighborhood-scale analysis. Then, Section 4.3 presents experimental results to investigate the observability ratio of the proposed corner-like feature detector, the viewpoint- and geometry-invariance properties of the extracted features, and the extraction of line segments in parallel with the proposed corner-like feature detector. Lastly, Section 4.4 presents a summary of this chapter, the main conclusions, and future work.

4.1 Background

Laser scanners allow building a dense representation of the environment as a 2D occupancy grid map, where each cell represents occupied, free, or unknown space [74]. Although grid maps can represent arbitrary objects, they are computationally intensive and require substantial memory [143]. An alternative method is to represent the environment as a set of features. In the case of 2D lasers, the features may include beacons[144], geometric cues (line [145–153] and curve [149–151, 154] segments, corners [145–151, 154–158], and free endpoints of segments [145, 150, 157, 158]), and semantic features such as parked vehicles [155, 159] and doors [146].

The correspondence between geometric features and the environment (*e.g.*, lines and corners) may be of interest for localization and mapping problems on mobile robots, as well as for user interaction with autonomous mobile systems. The current literature presents several algorithms for extracting line segments from 2D laser scan data [145–153], including an open-source implementation based on the ROS [96] framework[1]. As for corner detection, the most common methodology found in the literature is to extract corners from the intersection of line segments detected in the laser scan [145–148, 154–157, 159–161]. Another typical approach in the literature is to estimate curvature from the scan data [149–151, 158]. Next, both trends for extracting corner-like features are analyzed based on methods found in the literature.

4.1.1 Line fitting intersection

A trend in the literature is to extract corners at the intersections of lines detected in the 2D laser scan [145–148, 154–157, 159–161]. First, the laser data may be filtered to remove invalid values (zero range values [155]) and possible noise (*e.g.*, considering the minimum distance to the two neighbors of a certain point to remove isolated data [155], mean average filtering [157], or bilateral filters to remove non-linear noise [148]). Next, a segmentation step extracts sets of continuous points in the environment for line fitting estimation. Castellanos *et al.* [145] segment the scan into polygonal lines and subsequently divide them into segments using an iterative method. Falomir *et al.* [154] search for discontinuities in the data to form point subsets and use a recursive line-fitting algorithm [162] to obtain line segments. Fernández-Moral *et al.* [160] search for parameters of a 2D line that maximize the number of points supporting the line model using the RANdom SAmple Consensus (RANSAC) algorithm [163]. Mohamed *et al.* [147] use the line tracking algorithm [164] to iteratively segment lines in the scan and perform Principle Component Analysis (PCA) to compute the lines' parameters. Then, the intersection points between consecutive line segments define a corner feature in the environment.

Furthermore, Lin *et al.* [156] and Liu *et al.* [148, 157] use adaptive threshold segmentation to cluster range data into continuous spatial subsets. Although Lin *et al.* [156] state that an adaptive segmentation was used due to the non-uniform distribution of the 2D laser data, the authors do not specify the threshold used in their work. In Liu *et al.* [157] work, the threshold is based on the laser's angular resolution and the distance to the current point under analysis. Liu *et al.* [148] also establish a threshold depending on the angular resolution. Instead of considering only the distance of the point under analysis, the proposed approach [148] selects the larger distance between adjacent points. Even though Roumeliotis and Bekey [146] do not use a specific adaptive threshold, the proposed SEGMENTS algorithm with two-level EKF exhibits adaptive behavior. A stricter filter, in terms of the system's noise covariance, attempts to follow a straight line, while a more flexible EKF tries to follow a new segment during each scan. The four methods [146, 148, 156, 157] also define a corner feature as the intersection between consecutive line segments.

A specific application of corner detection with 2D laser data in the literature is vehicle detection [155, 159]. Jung *et al.* [155] cluster laser data using a fixed threshold between consecutive range values. Then, the method performs template matching for every point under analysis, assuming the point is the vertex of the L-shape of a vehicle in the scan. Jung *et al.* [155] use Singular Value Decomposition (SVD) to estimate the L-shape's line parameters and decompose the problem into line-curve estimation when the vehicle has a curved corner in the data. In contrast, Qu *et al.* [159] segment the data using Density Based Spatial Clustering of Applications with Noise (DBSCAN) [165]. The method performs L-shape fitting in two steps: searching for the L-shape vertices and corner-point localization, the latter based on RANSAC [163].

However, corner detection algorithms that rely on global search during line segment scanning may be limited to well-structured environments. Thus, we propose a corner-like detection method based on local line

[1] https://github.com/kam3k/laser_line_extraction

fitting, without requiring global line detection or recursive data search. Also, the experiments presented in Section 4.3 demonstrate the ability of the proposed detector to extract more points of interest from the scan in addition to environment corners.

4.1.2 Curvature-based detection

Other approaches to detecting corners in 2D laser data estimate the curvature. Núñez *et al.* [149] propose an adaptive curvature estimator that compares the k-nearest points on both sides of the point being evaluated. The k size is adjusted based on the distance between possible corners. A segmentation step precedes the curvature estimation to detect breakpoints using the adaptive breakpoint detector proposed by Borges and Aldon [166]. Then, Núñez *et al.* [149] extract corner, line, and curve segment features from the scan data by analyzing whether there are peaks in curvature, flat segments with zero average curvature, and flat segments with non-zero average curvature, respectively. Vázquez-Martín *et al.* [150] use the adaptive curvature estimator of Núñez *et al.* [149] to segment the scan into clusters of homogeneous curvature. Even though the existence of a corner is determined from the curvature function, its keypoint definition is based on the intersection of the two line features that generate the corner, to improve robustness to noise. Also, Certad *et al.* [158] modify the work of Núñez *et al.* [149] to consider two curvature estimations based on different noise-related thresholds, thereby improving the rate of true detection for corner and edge features in the experiments.

Moreover, Feng *et al.* [151] and Zhao *et al.* [161] report a correlation between the angle difference between the two fitted lines on both sides of a scan point and the curvature value. Feng *et al.* [151] introduce the micro-tangent line function as the variance between the tangents at two consecutive points in the scan. Similarly to Núñez *et al.* [149], line and curve segments are extracted from consecutive range readings with variance close to zero and with constant value larger than a threshold, respectively, and corner features are extracted from peaks in the variance function. Instead of only considering two consecutive points, Zhao *et al.* [161] define two distance-based regions on both sides of a point to estimate direction vectors and, consequently, the angle difference between those vectors. Candidate corner points are selected based on a fixed threshold for the angle difference. Then, the corner point is estimated via non-minimal suppression.

Still, most curvature-based methods for corner feature extraction associate the keypoint with a laser reading. Given the discrete, radial nature of 2D laser, the estimated corner may vary farther away. Also, there is the question of whether the curvature estimators are invariant to scale, angular resolution, and different viewpoints. We propose a detector based on the stability of intersection points from fitted lines at different distance-based scales. The experiments in Section 4.3 focus on evaluating the invariance of the proposed corner-like features.

4.2 Corner-Like Feature Detector

We propose a feature extraction algorithm on 2D laser data performing local line fitting to detect corner-like points in the scan. In summary, first, we transform the scan data from polar to Cartesian coordinates. This transformation maintains the data's angular coordinate order. For each point in the scan, our methodology considers multiple distance-based scales to extract arrival and departure neighborhoods, depending on whether the latter are in the negative or positive directions relative to the point's angular coordinate, respectively. Next, a linear regression algorithm fits lines to each neighborhood. The intersection point between the arrival and departure neighborhoods is a candidate feature keypoint at a given scale. Then, the following indicators are estimated to determine if the candidate keypoint is a corner-line feature or not: minimum number of points to fit a line, angle between arrival and departure lines, maximum RMSE allowed for the line fitting procedure, maximum distance of the candidate point to the intersection centroid on the multiple scales, and maximum distance to the laser point on analysis. The candidate keypoint is validated if it passes all previous indicators over a minimum number of consecutive scales, thereby improving robustness to noise in the laser data.

4.2.1 Linear least-squares regression

A generic definition for the 2D line model is presented in Equation (4.1), where (x,y) are the 2D point coordinates and γ_0, γ_1, and γ_2 are the model parameters. However, the solution for the generic definition estimated with a linear least-squares formulation from a set of 2D points would be a null vector.

$$\gamma_0 + \gamma_1 x + \gamma_2 y = 0 \tag{4.1}$$

An alternative formulation is to characterize the 2D line relative to one of the point coordinates, reducing from three to two parameters in the model. The 2D line formulation illustrated in Equation (4.2) is relative to the y coordinate with respect to x, where β_1 is the line slope and β_0 is the line intersection on the y-axis. The least-squares solution for the parameters β_1 and β_0 from a set of points $(x_i, y_i), i = 1, \ldots, N$ is formulated in Equation (4.3) and Equation (4.4), where $(\bar{x}, \bar{y})$ is the centroid point of the data [167]. Still, this model would not be valid when $\sum_{i=1}^{n}(x_i - \bar{x})^2 = 0$, given the indeterminate form caused in the parameter $\hat{\beta}_1$ (*e.g.*, all the data points with the same x coordinate and different y values).

$$y = \beta_0 + \beta_1 x \tag{4.2}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{4.3}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \tag{4.4}$$

The 2D line can be defined relative to x with respect to the y coordinate, as formulated in Equation (4.5). The parameters of this model are the line slope (α_1) and the line intersection on the x-axis (α_0). Nevertheless, this model has also a undetermined form when $\sum_{i=1}^{n}(y_i - \bar{y})^2 = 0$ (*e.g.*, all the data points with the same y coordinate and different x values).

$$x = \alpha_0 + \alpha_1 y \tag{4.5}$$

$$\hat{\alpha}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{4.6}$$

$$\hat{\alpha}_0 = \bar{x} - \hat{\alpha}_1 \bar{y} \tag{4.7}$$

Instead of only considering a single line model for data fitting, we select the best line model that fits the laser data. This selection is based on which model has the lower fitting RMSE, ε_{RMSE}. Equation (4.8) and Equation (4.9) represent the fitting RMSE of the 2D line models formulated in Equation (4.2) and Equation (4.5), respectively. The validation of the proposed selection approach for the line models is shown in Figure 4.1,

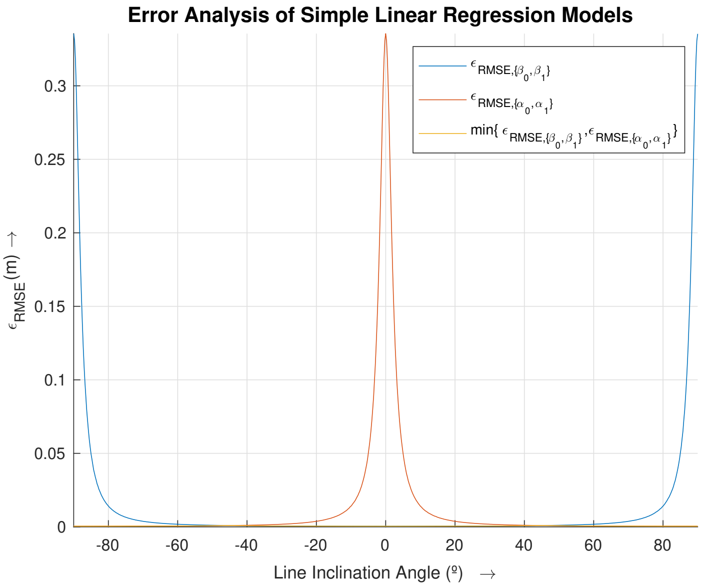


Figure 4.1: Error analysis of simple line regression models evaluating the fitting Root Mean Square Error (RMSE), ε_{RMSE}, for lines with an inclination angle between -90º and 90º and a simulation step of 0.25º, a set of 100 data points equally distributed along each line, a distance Gaussian noise added to the data point coordinates of zero mean and 0.03 m standard deviation, and 1000 simulation samples per line [66].

where selecting the model with the lowest RMSE achieves the best fitting line estimation.

$$\varepsilon_{RMSE,\{\hat{\beta}_0,\hat{\beta}_1\}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2} \tag{4.8}$$

$$\varepsilon_{RMSE,\{\hat{\alpha}_0,\hat{\alpha}_1\}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_i - \hat{\alpha}_0 - \hat{\alpha}_1 y_i\right)^2} \tag{4.9}$$

Other models and fitting algorithms could have been considered to fit a line to the 2D laser data, such as formulating a line as the vector to the normal of the infinite line and the position of the weighted mean of the contributing points [152], or using RANSAC to estimate the line parameters of the generic line model [163]. However, these models and fitting algorithms may require iterative-based optimization procedures and matrices inversions. In contrast, our methodology has the advantage of having closed-form equations for both the line parameters and the fitting error.

4.2.2 Feature extraction

In terms of extracting features from 2D laser data, first, we need to transform the scan data from polar to cartesian coordinates. This transformation preserves the data ordering by its angular coordinate to speed up the search for the closest consecutive laser points in the angular direction at a given distance-based neighborhood.

Next, we evaluate each of the laser points $(x_i, y_i), i = 1, \ldots, N$ as a possible feature keypoint. Figure 4.2 presents an example of the clustering and line fitting results following the proposed methodology on a laser point (x_j, y_j) at a distance scale $R_k = 0.15$ m, where this scale defines the distance-based neighborhood around the scan point. These results illustrate the clustering of the arrival $((x_i, y_i) \in \{(x_j, y_j)_{\text{arr},k}\})$ and departure $((x_i, y_i) \in \{(x_j, y_j)_{\text{dep},k}\})$ neighborhoods, representing the closest consecutive laser points at the negative and

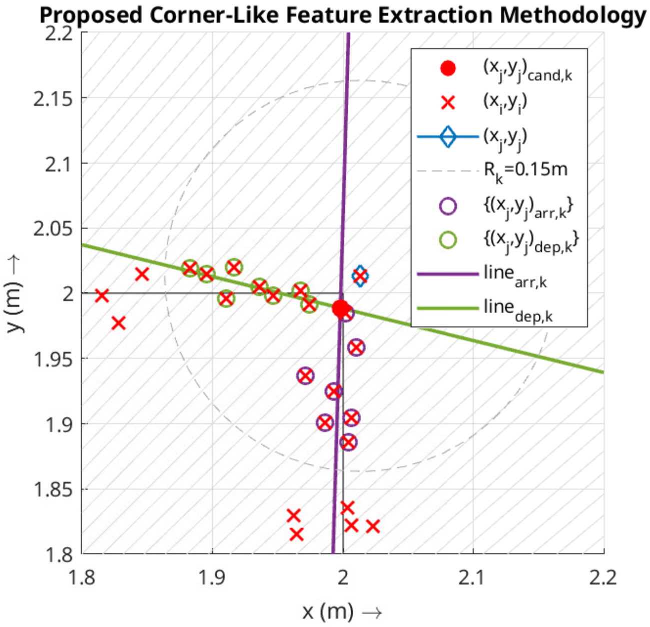


Figure 4.2: Example of the proposed corner-like feature extraction methodology on a laser point (x_j, y_j) at a distance scale $R_k = 0.15$ m. Scan points (x, y) simulated considering a 2D laser scanner with 0.25º angular resolution, 0.06–10.0 m range, and Gaussian noise with zero mean and 0.03 m standard deviation (same specifications as the Hokuyo UST-10LX sensor) [66].

positive sides of the point (x_j, y_j) in the laser angular direction, respectively. The intersection point between the fitted lines $line_{\text{arr},k}$ and $line_{\text{dep},k}$ is considered a candidature feature keypoint ($(x_j, y_j)_{\text{cand},k}$). This intersection point is the result of the lines intersection depending on which line model is selected for each neighborhood:

- $y = \beta_0^{\text{arr},k} + \beta_1^{\text{arr},k} x \; ; \; y = \beta_0^{\text{dep},k} + \beta_1^{\text{dep},k} x$

$$\begin{cases} x_{j,\text{cand},k} = \left(\hat{\beta}_0^{\text{dep},k} - \hat{\beta}_0^{\text{arr},k}\right) / \left(\hat{\beta}_1^{\text{arr},k} - \hat{\beta}_1^{\text{dep},k}\right) \\ y_{j,\text{cand},k} = \hat{\beta}_0^{\text{arr},k} + \hat{\beta}_1^{\text{arr},k} \cdot x_{j,\text{cand},k} \end{cases} \tag{4.10}$$

- $x = \alpha_0^{\text{arr},k} + \alpha_1^{\text{arr},k} y \; ; \; y = \beta_0^{\text{dep},k} + \beta_1^{\text{dep},k} x$

$$\begin{cases} y_{j,\text{cand},k} = \left(\hat{\alpha}_0^{\text{arr},k} \hat{\beta}_1^{\text{dep},k} + \hat{\beta}_0^{\text{dep},k}\right) / \left(1 - \hat{\alpha}_1^{\text{arr},k} \hat{\beta}_1^{\text{dep},k}\right) \\ x_{j,\text{cand},k} = \hat{\alpha}_0^{\text{arr},k} + \hat{\alpha}_1^{\text{arr},k} \cdot y_{j,\text{cand},k} \end{cases} \tag{4.11}$$

- $y = \beta_0^{\text{arr},k} + \beta_1^{\text{arr},k} x \; ; \; x = \alpha_0^{\text{dep},k} + \alpha_1^{\text{dep},k} y$

$$\begin{cases} y_{j,\text{cand},k} = \left(\hat{\alpha}_0^{\text{dep},k} \hat{\beta}_1^{\text{arr},k} + \hat{\beta}_0^{\text{arr},k}\right) / \left(1 - \hat{\alpha}_1^{\text{dep},k} \hat{\beta}_1^{\text{arr},k}\right) \\ x_{j,\text{cand},k} = \hat{\alpha}_0^{\text{dep},k} + \hat{\alpha}_1^{\text{dep},k} \cdot y_{j,\text{cand},k} \end{cases} \tag{4.12}$$

- $x = \alpha_0^{\text{arr},k} + \alpha_1^{\text{arr},k} y \; ; \; x = \alpha_0^{\text{dep},k} + \alpha_1^{\text{dep},k} y$

$$\begin{cases} y_{j,\text{cand},k} = \left(\hat{\alpha}_0^{\text{dep},k} - \hat{\alpha}_0^{\text{arr},k}\right) / \left(\hat{\alpha}_1^{\text{arr},k} - \hat{\alpha}_1^{\text{dep},k}\right) \\ x_{j,\text{cand},k} = \hat{\alpha}_0^{\text{arr},k} + \hat{\alpha}_1^{\text{arr},k} \cdot y_{j,\text{cand},k} \end{cases} \tag{4.13}$$

The following indicators determine if a candidate keypoint is a corner-like feature or not: minimum of 5 points in each neighborhood, an absolute angle between arrival and departure fitted lines in the range of $[45º, 135º]$, maximum ε_{RMSE} in line fitting of 0.025 m, maximum distance of the lines intersection points on the multiple scales to the intersection centroid of 0.025 m, and maximum distance of the intersection point to the scan point being evaluated of 0.05 m. The latter indicator avoids having features not representing physical points. Then, the candidate keypoint is validated if the point passes all previous indicators in a minimum number of 3 consecutive scales, to improve robustness to noise on the laser data. The scales considered in the experiments presented in this chapter are the following ones: $[0.05, 0.075, 0.1, 0.125, 0.15, 0.175, 0.2, 0.225, 0.25]$ m. However, the indicators' thresholds and the scale may be parametrized in run-time. The feature keypoint definition is the lines intersection centroid on the multiple scales.

4.3 Experimental Results and Discussion

The experiments were performed using a three-wheeled omnidirectional robot developed by the 5dpo robotics team [168]. This platform was adapted to have a 2D laser scanner, namely, the Hokuyo UST-10LX with 270º FOV, 0.25º angular resolution, and 40 Hz sampling rate. The sensor is aligned with the geometric center of the robot, and the wheel odometry parameters were estimated using the OptiOdom [57] calibration method. The implementation of the feature extraction algorithm is in C++ and ROS. The algorithm ran on a computer with a processor Intel Core i7-6700HQ with 8GB of RAM, and a Nvidia GPU GeForce GTX 960M.

In terms of the experimental methodology, first, a map of the scene was retrieved using the SLAM Toolbox [95] algorithm with a resolution of 0.03 m. Although this algorithm already estimates the robot pose relative to the map being created, this estimation may be improved if the whole map is used to localize the robot. So, the Adaptive Monte Carlo Localization (AMCL) [78] algorithm available in ROS was used to estimate the robot's localization relative to the map obtained from the SLAM process. With this localization estimation, the point clouds from the 2D laser scanner can be transformed to the map coordinate frame that is a static one, instead of being referenced to the moving laser frame. All parameters of the algorithms, launch configuration files, and scripts used in the experiments are also available in a public GitHub repository[2].

4.3.1 Feature extraction results

Figure 4.3 is an example of the feature extraction results. This example represents the feature keypoints $(x,y)^{t=t_i}$ at a time instant t_i, where the blue circle is the robot pose estimated by the AMCL [78] algorithm and the orange dotted lines represent the FOV of the sensor. The points $(x,y)^{\text{persist}}$ represent the locations of the 39 most persistent features in space extracted by the proposed methodology. These locations were found through manual selection of the features location considering their observability through the experiment (see Section 4.3.2). Further results relative to the features extraction with the proposed methodology are available in videos[3].

When analyzing visually the example results, the persistent features correspond not only to well-defined 90º corners but also to other corner-like points of interest. This observation is shown both in Figure 4.3 and in the videos[3], even detecting stable features in legs of passing people through the environment. As for the difference between the features detection at a certain time instant and the most persistent ones, this difference may be due to noise in the distance measures of the 2D laser.

4.3.2 Features observability

Next, the observability of the features extracted when moving the robot through the scene shown in Figure 4.3 was evaluated by processing the features location in the global map frame, using the experimental methodology explained previously. Figure 4.4 represents a color map of the features observability ratio, considering the

[2]https://github.com/INESCTEC/inesctec_mrdt_line_fit_based_corner_detector_2d_laser
[3]https://www.youtube.com/watch?v=desB9LX4hck&list=PLvp8fJUEPxYQjsuJAeiBVLEKuBU4Otu7e

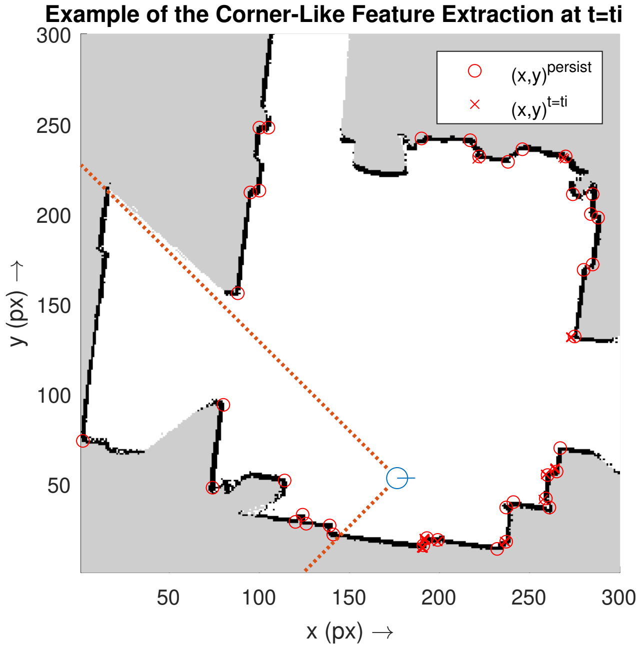


Figure 4.3: Example results for the proposed corner-like feature extraction methodology. Occupancy grid map obtained by using the SLAM Toolbox [95] algorithm, considering a map resolution of 0.03 m/px [66].

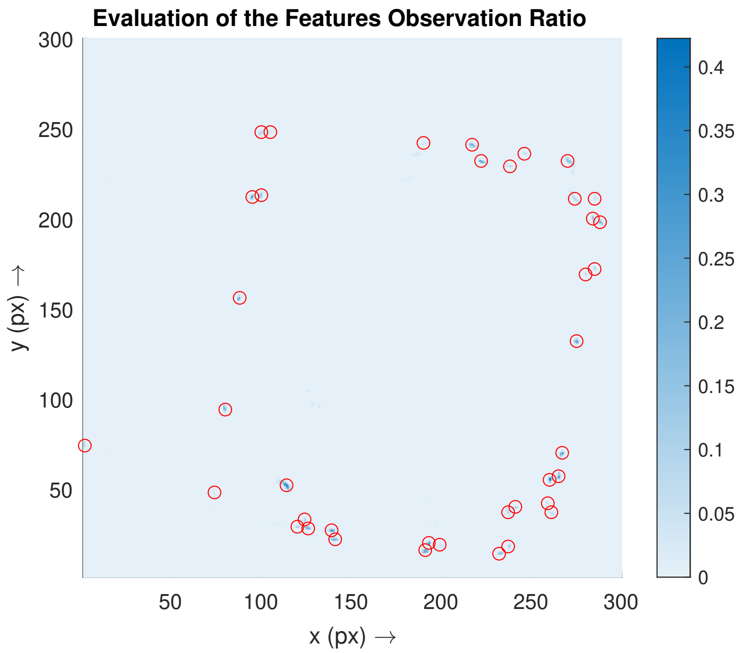
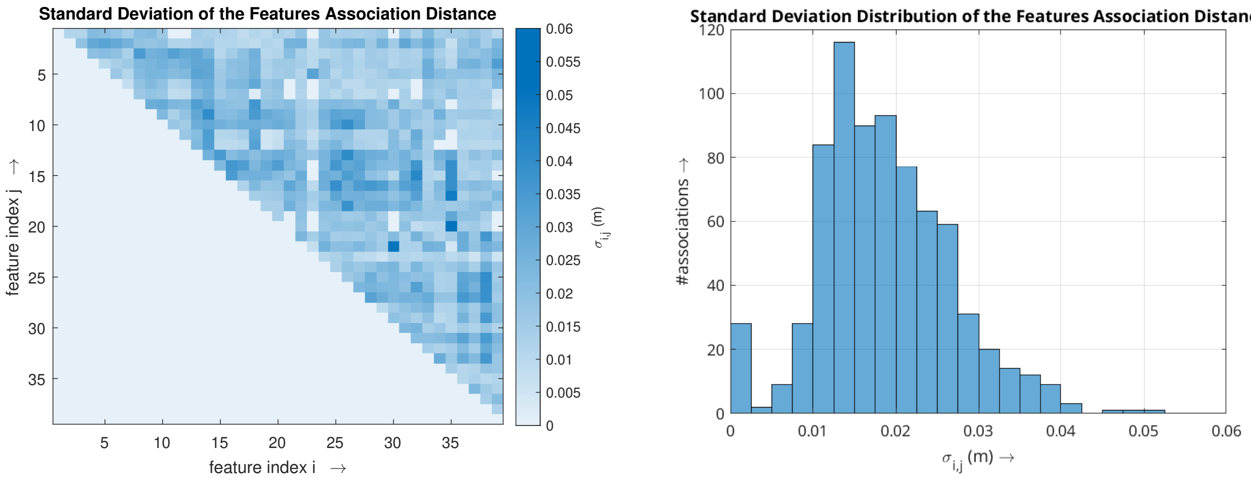


Figure 4.4: Evaluation of the corner-like features observation ratio, considering the robot pose along time, the Field-of-View (FOV) of the 2D laser scanner, and a map resolution of 0.03 m/px [66].

FOV of the sensor and the robot pose throughout time. This color map considers the same resolution of 0.03 m/px as the one used to localize the robot. Also, the observability map allowed the manually selection of the 39 persistent features also shown in Figure 4.3.

The first observation is the lower number of false positives. Other than the locations near the 39 most persistent features, the observability ratio of the other areas of the map is lower than 1.0%. Furthermore, there is a blur in the observability ratio near the location of the 39 persistent features, instead of being concentrated in



(a) (b)

Figure 4.5: Standard deviation analysis of the distance between corresponding corner-like features: (a) association matrix; (b) histogram distribution [66].

a single pixel. This blur is a consequence of the uncertainty of the location estimation of the robot, explaining why the maximum observability ratio when considering the resolution of 0.03 m/px of the map was only 42.23%. Indeed, if the observability analysis considers the N8-neighborhood of each pixel, this maximum observability ratio increases to 100% for 6 features. For the same consideration of the N8-neighborhood, 20 and 16 features have an observability ratio greater or equal to 50% and 75%, respectively. These results indicate the possible usefulness of the proposed corner-like feature detector for tasks such as localization and mapping using 2D lasers, given the stability of the feature keypoints.

4.3.3 Features association distance

Even though the analysis on the features observability indicate if the features are consistently seen in space, this evaluation is highly dependent on the location estimator required to transform the point cloud to a static frame. Another possible analysis in the context of feature-based localization systems is to evaluate the distance between different features throughout time, assessing their relative position instead of the absolute one in space. So, Figure 4.5 presents the analysis on the feature association distance in terms of standard deviation to evaluate the uncertainty on the relative position between the 39 most persistent features selected previously (see Figure 4.4). The estimation of the standard deviation only considers valid contributing measures, *i.e.*, only consider the distance between two features if both are visible in the same time instant. This estimation was possible by associating the features observable in each time instant to one of the 39 persistent ones if the error distance to the location manually selected is lower than 0.06 m. The latter threshold was needed to account both the unknown uncertainty of the location estimator and the noise on the laser measures (zero mean and 0.03 m standard deviation according to the official specifications for the Hokuyo UST-10LX sensor).

The maximum standard deviation of features association distance obtained in the experiments is 0.051 m. Although this value is greater than the one characteristic of the sensor, most of the associations have a non-zero standard deviation lower or equal to 0.03 m, representing 91.5% of the possible associations between features. Note that only non-zero values must be considered because, due to the sensor not having a 360º FOV and to possible physical obstructions on the laser view, not all associations are physically possible (in this case, 20 associations are not possible). Even when considering associations with a deviation lower or equal to 0.03 m, the proportion remains very significant at 59.3%. Thus, these results on the feature association distance underline the usefulness of the proposed corner-like features detector for localization processes, given that the geometric relations between features remain consistent and invariant.

4.3.4 Integration with a line-segment feature extraction algorithm

The final experiment in the scope of this chapter is shown in a video[4], where our corner-like feature extractor and an open-source line-segment feature extraction algorithm[1] run simultaneously. In most cases, there is a complementarity between our corner-like features and line-segment features. Indeed, there are situations in which no corners are detected while line features are present. At the same time, other situations in which the expected lines are too small and are therefore possibly rejected by the line extractor, leaving only corners to be detected. Although the situations shown in the video are also dependent on the environment surrounding the robot, this complementarity between different types of features is advantageous for localization and mapping processes, thereby improving the robustness and stability of the robot pose estimate by increasing the number of features observable in the environment. Still, the main goal of this experiment is to show that the proposed feature extraction algorithm may be run in parallel with other algorithms, leveraging the ROS [96] modular architecture of having independent nodes subscribing to the same data, in this case, the range data from the 2D laser scanner sensor.

4.4 Summary and Conclusions

This chapter addresses a feature extraction algorithm for corner-like features from 2D lasers data. Although the most common way to represent the environment from 2D lasers is via 2D occupancy grid maps, an alternative is to represent the scene as a set of features. These features may include beacons[144], geometric cues (line [145–153] and curve [149–151, 154] segments, corners [145–151, 154–158], and free endpoints of segments [145, 150, 157, 158]), and semantic features [146, 155, 159]. In the specific case of corner-like features, the most common approach in the literature is to intersect line-segment features to extract corners from 2D lasers data [145–148, 154–157, 159–161]. However, this methodology makes corner detection highly dependent on the line-detection algorithm and may be limited to well-structured environments. Another common approach is to estimate curvature from the scan data [149–151, 158]. Still, the influence of sensor noise on curvature estimation may lead to variant features and false positives.

Given the previous limitations, we propose a feature-extraction algorithm to detect points of interest in 2D laser scanners data that resemble physical corners. The proposed algorithm performs local line fitting at multiple distance scales around a given point, similar to Speeded Up Robust Features (SURF) [169]. These scales define arrival and departure distance-based neighborhoods around the points in the scan. The intersection of the fitted lines defines candidate feature keypoints. Then, a set of indicators based on line-fitting error, the angle between the arrival and departure lines, and consecutive observations of the same candidate keypoint across different scales defines a corner-like feature. Our methodology is independent of global search during line segment scanning, allowing other feature extraction methods to run concurrently.

[4] https://www.youtube.com/watch?v=7xMKtlvd0wU

Experimental results show that the proposed corner-like features correspond not only to well-defined 90º corners but may also detect other corner-like points of interest. More points of interest yield more features that may help stabilize the robot's localization when estimating its pose using the proposed corner-like features. Furthermore, we analyze the observability ratio of the corner-like features throughout the experiment, based on the robot pose and sensor FOV. When considering the robot pose uncertainty and sensor, 6 features had a 100% observability ratio, while 20 and 16 features had a ratio greater or equal to 50% and 75%, respectively. These results reveal viewpoint invariance of the proposed feature extractor algorithm. As for geometric invariance, when analyzing the association distance between the 39 most persistent features from the experiments, the maximum standard deviation of these distances for each pair of associated features was 0.051 m. In contrast, 91.5% of possible feature associations had a standard deviation of 0.03 m or less, which is equivalent to the sensor noise. These results demonstrate that geometric relations between corner-like features extracted through our methodology remain consistent and geometrically invariant.

Overall, our corner-like features demonstrate their usefulness for localization and mapping processes through their viewpoint- and geometry-invariant properties in the experiments. These features can be extracted simultaneously with line segments from 2D laser scanner data, and combining different feature modalities can provide complementary observations, thereby improving the robustness and stability of localization. A natural continuation of this work is the development of a feature-based 2D localization estimator for mobile robots using 2D laser scan data, which could later be integrated as a front-end within a SLAM framework. Although feature-based localization and mapping constitute a promising research direction, it also requires additional components, such as data association algorithms, uncertainty models for feature observations, possibly feature descriptors, and specific map data structures. These requirements are not commonly supported in ROS [96], in which the most widely adopted representation of a 2D environment is the occupancy grid map rather than a feature-based map. Therefore, the next chapters of this PhD Thesis focus on the localization and mapping of mobile robots using wheeled odometry and raw 2D laser scanner data, rather than relying on feature-based environment representations.

Chapter 5

Data Registration with Distance Maps

TBC

5.1 Background

TBC (ICP and its variants, etc.)

5.2 Distance Maps

TBC (unsigned distance map structure, types of distance map formulations – only NN, NN + distance, NN + 1st derivative, NN + 1st and 2nd derivative).

5.3 Data Registration

TBC

5.3.1 Least-squares on manifolds

TBC

5.3.2 Optimization algorithm

5.3.3 Covariance estimation

5.3.4 Problem formulation – point-to-point error formulation

TBC

5.3.4.1 State space

5.3.4.2 Measurement space

5.3.4.3 Prediction function

5.3.4.4 Error function

5.3.5 Problem formulation – point-to-plane error formulation

TBC

5.3.5.1 State space

5.3.5.2 Measurement space

5.3.5.3 Prediction function

5.3.5.4 Error function

5.4 Experimental Results and Discussion

TBC

5.4.1 Alignment results

5.4.2 Covariance estimation

5.4.3 Computation performance

5.5 Summary and Conclusions

TBC

Chapter 6

Simultaneous Localization and Mapping (SLAM) in Dynamic Environments

TBC

6.1 Background

TBC

6.2 Online SLAM

TBC

6.3 Dynamics Removal

TBC

6.4 Offline Global Bundle Adjustment

TBC

6.5 Experimental Results and Discussion

TBC

6.6 Summary and Conclusions

TBC

Chapter 7

Conclusions and Future Work

TBC

7.1 Conclusions and Contributions

TBC

7.2 Final Remarks and Future Work

7.2.1 Future Work

7.2.2 Involvement in the Scientific Community

TBC

7.2.2.1 Collaborations

Giorgio Sapienza, Rome (PhD Internship, 6 months, February - July 2024).

5dpo Robotics Team:

[170]

R.B. Sousa, P.G. Costa, and A.P. Moreira. "A pose control algorithm for omnidirectional robots". In: *2021 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*. Santa Maria da Feira, Portugal, 2021, pp. 91–96. DOI: 10.1109/ICARSC52212.2021.9429803

[171]

R.B. Sousa, C.D. Rocha, J.G. Martins, J.P. Costa, J.T. Padrão, J.M. Sarmento, J.P. Carvalho, M.S. Lopes, P.G. Costa, and A.P. Moreira. "A robotic framework for the robot@factory 4.0 competition". In: *2024 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*. Paredes de Coura, Portugal, 2024, pp. 66–73. DOI: 10.1109/ICARSC61747.2024.10535935

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M.S. Lopes, J.D. Ribeiro, A.P. Moreira, C.D. Rocha, J.G. Martins, J.M. Sarmento, J.P. Carvalho, P.G. Costa, and R.B. Sousa. "From competition to classroom: a hands-on approach to robotics learning". In: *2025 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*. Funchal, Portugal, 2025, pp. 170–176. DOI: 10.1109/ICARSC65809.2025.10970153

7.2.2.2 Scientific projects

APPLICATION: SLAM4all (SLAM em tempo real para multi-domínio com múltiplos robôs sob transmissão de dados limitada).

PROJECTS: VINCI7D (Gerber parser); GreenAuto; PRODUTECH R3.

- PRODUTECH R3:

[173]

T.B. Levin, J.M. Oliveira, R.B. Sousa, M.F. Silva, B.S. Parreira, H.M. Sobreira, and H.S. Mendonça. "Image and command transmission over the 5G network for teleoperation of mobile robots". In: *2024 7th Iberian Robotics Conference (ROBOT)*. Madrid, Spain, 2024, pp. 1–8. DOI: 10.1109/ROBOT61475.2024.10797434

[174]

D. Caldana, A. Cordeiro, J.P. Souza, R.B. Sousa, P.M. Rebelo, A.J. Silva, and M.F. Silva. "Pallet and pocket detection based on deep learning techniques". In: *2024 7th Iberian Robotics Conference (ROBOT)*. Madrid, Spain, 2024, pp. 1–8. DOI: 10.1109/ROBOT61475.2024.10797410

[175]

F.D. Pacheco, P.M. Rebelo, R.B. Sousa, M.F. Silva, and H.S. Mendonça. "Integrated RFID system for intralogistics operations with industrial mobile robots". In: *2025 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*. Funchal, Portugal, 2025, pp. 124–131. DOI: 10.1109/ICARSC65809.2025.10970172

- VINCI 7D:

[176]

R.B. Sousa, C. Rocha, H.S. Mendonça, A.P. Moreira, and M.F. Silva. "Gerber file parsing for conversion to bitmap image–the VINCI7D case study". In: *IEEE Access* 10 (2022), pp. 69659–69679. DOI: 10.1109/ACCESS.2022.3187042

7.2.2.3 Supervision

B.Sc. supervision (not official) at FEUP (February - June 2023; 2nd semester 2022/23): Rodrigo de Vasconcelos e Miguel (Compensação do Movimento do Robô nos Dados de Sensores Exteroceptivos); José Miguel Guedes (Controlo de Trajetórias de Robôs Móveis com o Hangfa Discovery Q2); André Luís Pires (Robótica e Sistemas Autónomos em Localização e Mapeamento de Robôs Móveis com o 3-omni 5DPO MSL); Ana Luísa Mota (Robótica e Sistemas Autónomos em Localização e Mapeamento de Robôs Móveis com o Hangfa Discovery Q2).

B.Sc. supervision (not official) at ISEP (March - July 2023; 2nd semester 2022/23): Paulo Ricardo Barbosa (Teste de algoritmos de mapeamento open source recorrendo ao sensor RS-Helios-5515).

BII at INESC TEC: Daniele da Mota Caldana (Aplicação para deteção de local para manipulação de paletes e paletes danificadas); Duarte Filipe Monteiro (Caixas metalicas GreenAuto); David José Castro (Caixas metalicas GreenAuto); João Miguel Oliveira (tele-controlo AMRs); Thiago Baldassarri Levin (comunicações 5G).

References

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[2] R. Siegwart, I.R. Nourbakhsh, and D. Scaramuzza. *Introduction to autonomous mobile robots*. Ed. by R.C. Arkin. Second. The MIT Press, 2004. ISBN: 978-0-262-01535-6.

[3] C. Cadena, L. Carlone, H. Carrillo, Y. Latif, D. Scaramuzza, J. Neira, I. Reid, and J.J. Leonard. "Past, present, and future of simultaneous localization and mapping: toward the robust-perception age". In: *IEEE Transactions on Robotics* 32.6 (2016), pp. 1309–1332. DOI: 10.1109/TRO.2016.2624754.

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Appendix A

A Systematic Literature Review on Long-Term Localization and Mapping for Mobile Robots

[65] R.B. Sousa, H.M. Sobreira, and A.P. Moreira. "A systematic literature review on long-term localization and mapping for mobile robots". In: *Journal of Field Robotics* 40.5 (2023), pp. 1245–1322. DOI: 10.1002/rob.22170.

Appendix B

Integrating Multimodal Perception into Ground Mobile Robots

[67] R.B. Sousa, H.M. Sobreira, J.G. Martins, P.G. Costa, M.F. Silva, and A.P. Moreira. “Integrating multimodal perception into ground mobile robots”. In: *2025 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*. Funchal, Portugal, 2025, pp. 104–111. DOI: 10.1109/ICARSC65809.2025.10970176.

Appendix C

Line Fitting-Based Corner-Like Detector for 2D Laser Scanners Data

[66] R.B. Sousa, H.M. Sobreira, M.F. Silva, and A.P. Moreira. "Line fitting-based corner-like detector for 2D laser scanners data". In: *2024 10th International Conference on Automation, Robotics and Applications (ICARA)*. Athens, Greece, 2024, pp. 210–216. DOI: 10.1109/ICARA60736.2024.10553006.